



UNIVERSITI TEKNOLOGI MARA (UiTM)

FACULTY OF INFORMATION SCIENCE

**BACHELOR OF INFORMATION SCIENCE (HONS) INFORMATION SYSTEMS
MANAGEMENT (CDIM262)**

IMS552 - INFORMATION SYSTEMS MANAGEMENT

GROUP ASSIGNMENT:

POS MALAYSIA BERHAD

PREPARED BY:

MUHAMMAD ZAHID HANAFI BIN MOHAMAD AZIAN	2025127713
NUR 'AAINII NABILAH BINTI MOHAMAD NASIR	2025119807
NUR AISYAH BINTI ROSLI	2025180601
NURUL AIMI UMAIROH BINTI ZULKARIA	2025162077

PREPARED FOR:

DR HASNAH BINTI HASHIM

GROUP:

CDIM262 5C

SUBMISSION DATE:

WEEK 12

GROUP ASSIGNMENT:

POS MALAYSIA BERHAD

BY:

MUHAMMAD ZAHID HANAFI BIN MOHAMAD AZIAN

NUR 'AAINII NABILAH BINTI MOHAMAD NASIR

NUR AISYAH BINTI ROSLI

NURUL AIMI UMAIROH BINTI ZULKARIA

UNIVERSITI UNIVERSITI TEKNOLOGI MARA (UiTM)

KAMPUS PUNCAK PERDANA SHAH ALAM

FACULTY OF INFORMATION SCIENCE



STUDENT PLEDGE OF ACADEMIC INTEGRITY

As a student of Universiti Teknologi MARA (UiTM), it is our responsibility to act in accordance with UiTM's academic assessment and evaluation policy. We hereby pledge to act and uphold academic integrity and pursue scholarly activities in UiTM with honesty and responsible manner. We will not engage or tolerate acts of academic dishonesty, academic misconduct, or academic fraud including but not limited to:

- a. **Cheating:** Using or attempt to use any unauthorized device, assistance, sources, practice or materials while completing academic assessments. This include but not limited to copying from another, allowing others to copy, unauthorized collaboration on an assignment or open book tests, or engaging in any act or conduct that can be construed as cheating.
- b. **Plagiarism:** Using or attempts to use the work of others (ideas, design, words, art, music, etc.) without acknowledging the source; using or purchasing materials prepared by another person or agency or engaging in other behavior that a reasonable person would consider as plagiarism.
- c. **Fabrication:** Falsifying data, information, or citations in any academic assessment and evaluation.
- d. **Deception:** Providing false information with intend to deceive an instructor concerning any academic assessment and evaluation.
- e. **Furnishing false information:** Providing false information or false representation to any UiTM official, instructor, or office.

With this pledge, I am fully aware that I am obliged to conduct myself with utmost honesty and integrity. I fully understand that disciplinary action can be taken against me if I, in any way, violate this pledge.

Name & Matric Number:

MUHAMMAD ZAHID HANAFI BIN MOHAMAD AZIAN	2025127713
NUR 'AAINII NABILAH BINTI MOHAMAD NASIR	2025119807
NUR AISYAH BINTI ROSLI	2025180601
NURUL AIMI UMAIROH BINTI ZULKARIA	2025162077

Course Code: IMS552

Programme Code: CDIM262

Faculty / Campus: UiTM Puncak Perdana

ABSTRACT

ACKNOWLEDGEMENT

In the name of Allah, the Most Gracious, the Most Merciful. we would like to express my sincere gratitude to the Almighty for providing me with the strength, patience, and motivation to complete this group assignment. Although we faced some difficulties during the process, we are truly thankful for the ability to overcome them and finish this task successfully.

Firstly, we would like to give my highest appreciation to my lecturer, Dr Hasnah Binti Hashim, for giving me the opportunity to work on this assignment. Her clear instructions, helpful feedback, and relevant examples helped me better understand the topic and complete the assignment with confidence. Her continuous support and encouragement made a positive impact on my learning experience. We would also like to thank my classmates who offered guidance and shared useful opinions during our discussions. Even though the project is an individual task, their willingness to share knowledge and ideas helped me improve my understanding and inspired me to do better.

In addition, we are thankful for the resources provided by the university, including reference materials and online databases, which were very advantageous during the research process. These resources allowed us to gather information and support our ideas with appropriate evidence.

Finally, we would like to thank my parents for their constant support throughout this assignment. Their encouragement, advice, and financial assistance were essential to me. Their belief in my abilities gave us the motivation to keep working and complete this task on time. Overall, we are very grateful to everyone who contributed directly or indirectly to the successful completion of this assignment.

TABLE OF CONTENTS

ABSTRACT	4
ACKNOWLEDGEMENT	5
1.0 INTRODUCTION OF PROJECT	9
1.1 Introduction.....	9
2.0 ORGANIZATION BACKGROUND	10
2.2 History of the Organization.....	11
2.3 Core Business Activities	12
2.4 Corporate Vision, Mission, Objectives and Goals.....	14
2.6 Functional Description of Departments/Divisions/Units	17
2.7 Roles and Functions of Departments/Divisions/Units	18
3.0 DATA USED IN THE ORGANIZATION	20
3.1 Customer Data	20
3.2 Parcel and Logistics Data.....	20
3.3 Financial Data.....	21
3.4 Employee Data	21
3.5 Operational Data.....	21
4.0 LIST OF INFORMATION SYSTEMS	22
4.1 List of Information Systems Used in the Organization	22
4.2 Brief Explanation, Functions and Objectives of Each Information System.....	23
4.2.1 Enterprise Resource Planning (ERP) System – SAP.....	23
4.2.2 Billing System.....	23
4.2.3 Courier Management System.....	23
4.2.4 Customer Relationship Management (CRM) System	24
4.2.5 Big Data Analytics System	24
4.2.6 Point of Sale (POS) System.....	24
5.0 TYPES OF INFORMATION SYSTEMS.....	25
5.2 Billing System	26
5.2.1 System Background	26
5.2.2 Functions and Purpose	27
5.2.3 Classification of the Billing System.....	27

5.3 Courier Management System	27
5.3.1 System Background	27
5.3.2 Functions and Purpose	28
5.3.3 Classification of the Courier Management System	28
5.4 Customer Relationship Management (CRM).....	29
5.4.1 System Background	29
5.4.2 Functions and Purpose	29
5.4.3 Classification of the CRM System	29
5.5 Big Data Analytics System	30
5.5.1 System Background	30
5.5.2 Functions and Purpose	30
5.5.3 Classification of the Big Data Analytics System	30
5.6 Point of Sale (POS) System	31
5.6.1 System Background	31
5.6.2 Functions and Purpose	31
5.6.3 Classification of the POS System	31
6.0 DETAILS OF SAP ENTERPRISE RESOURCE PLANNING (ERP)	32
6.2.2 Financial Management	34
7.0 INFORMATION SYSTEM COMPONENTS	37
8.0 INFORMATION SYSTEM ACTIVITIES	46
9.0 HARDWARE AND SOFTWARE CONFIGURATION	55
9.1 Hardware Configuration.....	55
9.2 Software Configuration.....	56
10.0 INFORMATION SYSTEM PROBLEMS	57
10.1 Current Issues and Challenges.....	57
10.2 Causes of the Problems	58
10.3 Impact on Organizational Performance	59
11.0 PERFORMANCE OF THE INFORMATION SYSTEM	59
11.1 System Efficiency	59
11.2 System Effectiveness.....	60
11.3 User Satisfaction	60
11.4 Contribution to Organizational Operations	61

12.0 SWOT ANALYSIS OF THE INFORMATION SYSTEM	62
13.0 GENERATE AND EVALUATE ALTERNATIVES	65
14.0 RECOMMENDATIONS AND ACTION PLAN.....	67
15.0 CONCLUSION	70
REFERENCES	71
APPENDICES	72

1.0 INTRODUCTION OF PROJECT

1.1 Introduction

This project was prepared as part of the requirements of the course IMS552 Information Systems Management. The purpose of this study is to get actual knowledge and understanding of how information systems are managed and used in real organisational settings. To achieve this objective, an interview session was conducted with Pos Malaysia Berhad which is the one of the leading postal and logistics service providers in Malaysia.

Firstly, modern organisations are heavily dependent on information systems. They enable organisations to collect, process, store and transmit information efficiently, to facilitate daily tasks and decision making. Information technology in the digital corporate world of today is used to promote efficiency, improve quality of service and remain competitive. Thus, it is necessary for students of Information Systems Management to understand how information systems are implemented and managed in an organization.

Furthermore, the interview with Pos Malaysia Berhad provides a lot of information on the organizations functioning, departments and information systems. The interview was conducted to obtain information on the systems employed by the organization, their role, components and how they contributed to organisational success. The study also examines the role of information technology in improving internal business communication, operations, customer service and management of the corporation.

Moreover, this project presents additionally a chance to distinguish between theoretical material learned in the classroom and practical applications in the industry. The observable and quantifiable notions are information systems management, organisational structure and business processes via the actual practices employed by Pos Malaysia Berhad. This exposure allows us to get a deeper insight into how technology is used by organisations to meet their aims and objectives.

Lastly, the evaluation also evaluates the effectiveness of the information system selected by the organisation. This analysis includes identifying the strengths and weaknesses of the system, assessing its performance and investigating potential areas for improvement. These traits are important to grasp, because businesses always have challenges in terms of technological improvements, consumer expectations, data management and operational performance.

1.2 Objectives of the Study

- To understand the background of the organisation, business operations and operational structure of Pos Malaysia Berhad.
- To investigate the information system used in Pos Malaysia Berhad for everyday operation and business activities.
- To analyse the functions, elements and operations of information system selected which is implemented in the organization.
- To identify the strengths, weaknesses, opportunities and threats (SWOT) of the chosen information system.
- To propose suitable recommendations and improvement plans to increase the effectiveness and performance of the information system.

2.0 ORGANIZATION BACKGROUND

2.1 Company Background

Pos Malaysia Berhad is the national postal and logistics services provider in Malaysia. The company provides mail, parcel and other logistical services across the country. Pos Malaysia operates a wide range of post offices, postal agents, processing facilities and delivery hubs to serve millions of customers ranging from individuals to corporate and government entities. The corporation is a well-established nationwide company. It is one of the most trusted and famous service providers in Malaysia.

In addition, the logistics and postal industry are a key component of the economy that enables growth and communication in the current fast paced corporate environment. Consequently, Pos Malaysia has been constantly changing their company to stay competitive and relevant. The company has grown from a typical letter delivery company. Instead, it has diversified into courier services, logistics solutions, warehousing, fulfilment services, e-commerce support and digital solutions. This change reflects the companies pledge to stay ahead of the technology curve and adapt to its customer's needs.

Moreover, Pos Malaysia also plays a significant role in helping the expansion of Malaysias e-commerce industry. With the increased popularity of online shopping, demand for parcel delivery and logistics services has seen a huge rise. To meet these objectives, Pos Malaysia has invested in modern technology, automated systems and digital platforms to enhance operational efficiency and the customer experience. Pos Laju services will provide customers with a more reliable and faster delivery service of parcels locally and overseas.

Furthermore, the interview shows the digital technology and information systems are the main component of Pos Malaysia in the everyday operations. The corporation has a number of information systems. These systems are used to manage the customer information, package tracking, delivery scheduling, operational monitoring and business reporting. Using these solutions a corporation can do many transactions without affecting the quality of service and operational precision. The information system facilitates the coordination of departments and gives management with the opportunity to make sound business decisions.

Besides that, Pos Malaysia likewise places a strong emphasis on customer satisfaction and service excellence. The organization is continually looking for methods to improve the quality of its service by deploying innovative solutions and streamlining its operational processes. The digital transformation initiatives allow our clients to take advantage of online services, monitor their goods in real time and get information on their deliveries. The initiatives are intended to enhance transparency and convenience for consumers and strengthen the company's competitiveness in the logistics field.

2.2 History of the Organization

The roots of Pos Malaysia Berhad date back to the early 1800s when postal services began being provided in the British-colonial Straits Settlements of Penang, Malacca, and Singapore. These early services provided merely a communication channel for letters, documents, and newspapers, servicing both the colonial government and the general population. After the independence of Malaysia in 1957, the government continued to run these services through the Jabatan Perkhidmatan Pos Tanah Melayu, which was a government agency running all national postal operations.

One of the key milestones of the organisation's history happened on 1 January 1992 when the Malaysian Postal Services Department became a corporation named Pos Malaysia Berhad under the Postal Services (Successor Company) Act 1991. Corporatisation meant an absolute shift from a government department to a commercially oriented organisation, when all assets and liabilities related to postal services were passed to the new corporation. The idea behind the corporatisation was improving the efficiency and performance of the postal service by applying the business-like approach to its operations.

On 28 August 2001, Pos Malaysia listed on the Kuala Lumpur Stock Exchange via a reverse takeover of Phileo Allied Berhad, switching itself from a Government-Linked Company (GLC) to a public listed company. In 2007, Pos Malaysia received its current name Pos Malaysia Berhad, changing from its previous one of Pos Malaysia and Services Holdings Berhad. One more restructuring happened in 2011 when DRB-HICOM Berhad acquired Khazanah Nasional Berhad's 32% equity in the company, thus placing Pos Malaysia in the list of companies belonging to the DRB-HICOM group. In 2016, DRB-HICOM increased their shareholding in Pos Malaysia.

In 2021, Pos Malaysia entered an important stage of transformation journey, launching what could be called internally its Phase 1 of transformation named "Back to Black". This stage focused on solving basic operational and financial problems, such as cost management and process optimisation, to stabilise the company financially amidst falling volume of traditional mail. Phase 2 of the transformation, which was called "Transform the Core" and followed Phase 1, entailed even deeper technology upgrades and cultural shift towards innovation and operational excellence. One of the actions taken during this transformation was cloud migration of Pos Malaysia to Amazon Web Services in 2022, which was done with the help of Minfy Technologies and AWS Professional Services.

Today, Pos Malaysia Berhad is a national postal and logistics services provider in Malaysia, having the exclusive mandate under the universal service obligation to provide postal services in the country according to the Postal Services Act 2012. Having a history of more than 200 years, the company transformed from a simple letter delivering company into a diversified group of businesses, whose segments cover postal, logistics, aviation, and others. The company services more than 11 million addresses across Malaysia through a network of more than 3,500 touchpoints (Pos Malaysia Berhad, 2023).

2.3 Core Business Activities

There are four major segments of the business of Pos Malaysia Berhad. Each of them caters to specific types of customers and operations. Together, these segments provided a total group revenue of RM1.87 billion for the fiscal year 2023.

Postal is the biggest segment of the group's business, providing RM1,086.11 million or 58.1% of the total group revenue in 2023. It consists of domestic mail and parcel delivery with the help of such well-known brands as Pos Mel and Pos Laju, international mail and parcel services, retail services through post office counters, including bill payments and licence renewals, and different custom solutions, such as mail room management and direct mail

services. Pos Malaysia Berhad is the only universal mail services provider in Malaysia according to the national mandate and thus provides its services both in urban and rural communities that may not be serviced effectively by any private courier.

Logistics is the second largest segment of the group's business, providing RM312.31 million or 16.7% of the total revenue. This segment consists of a variety of supply chain and freight management activities, which include project logistics, e-commerce logistics, vehicle shipping, haulage services, warehousing and distribution, as well as air and sea freight services. This segment is mainly operated by Pos Logistics Berhad and its subsidiaries. Taking into consideration the growth of e-commerce in Malaysia, it plays a very important role in the strategy of the group's development.

Aviation is the third major segment of the group's business, providing RM303.50 million or 16.2% of the total revenue. It is operated by Pos Aviation Sdn. Bhd. and consists of ground handling, cargo handling, in-flight catering, aircraft maintenance and engineering services, as well as meet-and-greet airport services. During the fiscal year of 2023, the Aviation segment provided approximately 3,680,478 in-flight catering meals and handled air cargo logistics at major airports in Malaysia.

Others is the smallest among the major segments of the group's business, providing RM168.83 million or 9.0% of the total revenue. This segment consists of several diversified subsidiaries of the company. Pos Ar-Rahnu is engaged in providing Islamic microfinancing and gold centre services. Pos Digicert is a licensed digital certificate authority that provides digital identity creation services. Datapos is engaged in mailing solutions, including data processing, enveloping, bulk mail, and delivery services (Pos Malaysia Berhad, 2023).

2.4 Corporate Vision, Mission, Objectives and Goals

Being a national operator with over 200 years of history, Pos Malaysia Berhad has a well-formulated corporate direction that sets the strategy, culture, and daily operations of the company. In the table below, all the elements of the corporate direction are represented from the company's overall vision to measurable goals based on the information from the Annual Report 2023 (Pos Malaysia Berhad, 2023).

Category	Statement	Description
Vision	<i>"We Deliver. We Connect. We Improve Lives."</i>	This is the long-term vision of Pos Malaysia. It shows what the company strives to become and what promise it gives to its clients. The vision shows the ambition of the company to become much more than a postal provider and to make the world of its clients richer.
Mission	<i>"Passionate about building trust to connect lives and businesses for a better tomorrow."</i>	This is the mission of Pos Malaysia that explains what the organisation is about and what it is supposed to do in order to reach its vision. It is based on trust and connectivity and puts the focus on the positive impact that Pos Malaysia makes on the community.
Core Values	<i>Own It, Build Trust, One Team, Drive Innovation, Move Fast, Delight Customers</i>	These six values define the desired conduct of the employees. They serve as guidelines for the culture and actions within the company and its interactions with customers and clients. For instance, Own It encourages the culture of accountability, whereas Drive Innovation stimulates to challenge existing processes and find solutions.

Objectives	<i>POS 2.0 Transformation Strategy</i>	Strategic objectives of Pos Malaysia are defined by three main pillars: financial sustainability, operational excellence, and continuous innovation. Some of the objectives are: to grow the revenues from parcels and logistics to compensate for the decline in mail volume, increase the effectiveness of last-mile delivery, increase digital capacity with the help of cloud and data analytics, and enhance the customer experience via omnichannel platforms.
Goals	<i>Measurable targets for service and sustainability</i>	Measurable goals of Pos Malaysia include delivery to more than 11 million addresses across Malaysia, high delivery success rate, the introduction of automation and AI in customer service and logistics operation and achieving the carbon emission reduction targets.

It is necessary to note that the Vision and Mission of Pos Malaysia complement each other in terms of defining the identity of the organisation. While the vision shows what Pos Malaysia is aspiring to do, the mission reveals its inner purpose and value-based approach to achieving its vision. Objectives and goals translate these aspirations into strategies and measurable results that are being assessed annually (Pos Malaysia Berhad, 2023).

2.5 Organization Structure Chart

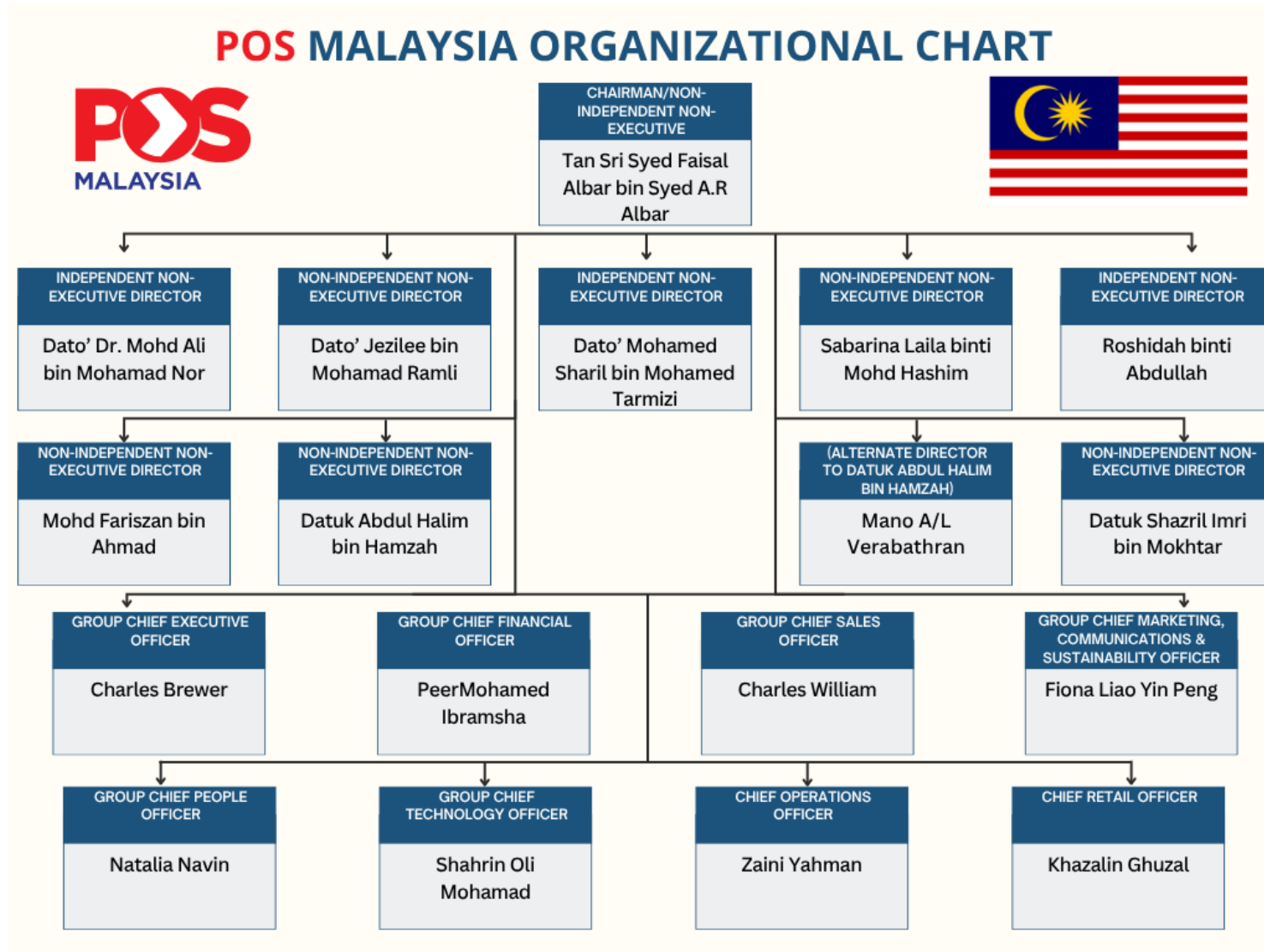


Figure 2.5.1: Organizational Chart of Pos Malaysia

2.6 Functional Description of Departments/Divisions/Units

From the interview with Pos Malaysia Berhad, it was found that the organisation has several important functional departments which cooperate in achieving its business activities. The various departments are interlinked with integrated information systems, especially the SAP ERP system, which links various departments and allows data to be managed centrally.

DEPARTMENT	DESCRIPTION
Human Resource (HR) Department	The HR department looks after all kinds of issues concerning workforce management in the organization. Its primary activities are recruitment and separation of employees, training and development activities, attendance and overtime, and employee benefit administration – EPF contributions, tax deductions, medical claims, allowances, etc. HR also oversees staff development programmes as per the human capital development goals. The HR module in the ERP system is used to manage payroll, track employee attendance and leave, and manage employee benefits.
Finance Department	Finance is responsible for the financial wellbeing and reporting needs of the organisation. Its duties include billing and invoice creation, account receivables management, cost control, payroll processing (in collaboration with HR), general ledger management and tax and payments processing. The SAP ERP system processes payroll for roughly 15,000 people, generates about 100,000 invoices, and stores about 11 terabytes of financial and operational data in the SAP ERP system database.
IT / Systems Department	The IT department has the responsibility of managing the technical infrastructure to support all information systems within the organisation. This involves hardware, servers, network architecture, system monitoring and activities of maintenance. The interview revealed a team of about eight individuals, including internal staff, external vendors and contractors, involved in the administration, monitoring and tech support for the SAP ERP system.

Operations / Logistics Department	Operations department oversees the company's basic courier and delivery operations, such as parcel processing, delivery planning, and logistics coordination. This department is supported by a courier management system and the Point of Sale (POS) system, which run transactional and movement data throughout the delivery network.
Customer Service / CRM Department	This department is responsible for customer enquiries, feedback management and relationship management activities. It integrates seamlessly with the Customer Relationship Management (CRM) system, which holds customer interaction logs information and gives data for enhancing the expertise.

2.7 Roles and Functions of Departments/Divisions/Units

All the departments in Pos Malaysia Berhad operate in an integrated way, where each department plays a part in achieving the overall mission of efficient postal, logistics and financial services delivery. The HR department takes care of the entire lifecycle of employees, from recruitment to separation, and manages their payroll, benefits, and tax-related matters accurately and timely. The HR department can use the SAP ERP system to process payroll data for thousands of employees, create Electronic Assessment (EA) forms that are needed for submitting tax-related information to the government and maintain time attendance data for overtime/leave calculation.

The Finance department is integral in ensuring the financial accuracy and accountability of the organisation. The Finance team operates within the SAP ERP system to produce thousands of invoices each month, maintain general ledger accounts, cost centres and generate management reports to assist in strategic decision-making. The seamless integration between the Finance module and HR module in SAP guarantees that payroll information is exact and transferred without a hitch to the monetary accounts, improving data reliability and minimizing manual mistakes between the two divisions.

The IT department is the backbone of the information systems infrastructure of the organisation. The team managing the SAP ERP system from the interview is relatively small, with around 8 people (vendors and contractors). Although the team is small, they monitor system performance through CPU usage tracking, memory utilisation, IOPS (input/output operations per second) and query response time measurement tools. The team provides High

Availability (HA) configurations, backups and backups restoration and interfaces with SAP vendors for licensing, updates and technical support to maintain system availability.

The Operations department manages the seamless flow of parcels and mail through the delivery system, ensuring that parcels are tracked, scheduled and operational reports are generated through the courier management system. The Customer Service department leverages the CRM system to handle customer profiles, document customer service interactions, and analyse customer feedback, helping to inform decisions regarding service enhancements. These departments are interdependent in terms of the flow of data, and the data is used for reporting, service provision and strategic decision making at Pos Malaysia Berhad.

3.0 DATA USED IN THE ORGANIZATION

According to the interview done with the representative from Pos Malaysia Berhad, the organisation has to deal with various types of data in its different business activities. These data categories play a pivotal role in the day-to-day functioning of the enterprise's information systems, especially the SAP ERP system that holds and deals with a vast amount of structured information for financial reporting, human resource management, and enterprise operations. The company's overall ERP database was reported to be around 11 terabytes, a testament to the data that the system stores continuously.

3.1 Customer Data

Customer data refers to any information about the customers or businesses that utilize Pos Malaysia's services. This involves customer names, contact information, billing addresses, service subscription information, account histories, and transaction information. The data for customers come mainly from the counter transactions carried out in the post offices, as well as from orders placed online via the company's electronic platforms and records of parcel deliveries. The CRM system is especially applicable to customer data management, where it stores interaction data and helps customer service teams handle customer inquiries and provide good customer service. The SAP ERP system is also used for customer billing, where the details of the invoices issued and the payment history are stored for financial purposes.

3.2 Parcel and Logistics Data

Parcel and logistics data is data on the tracking and movement of all mail items and parcels handled by Pos Malaysia. This includes shipping tracking numbers, parcel size and weight, shipping routes, shipping times, sender and receiver details, delivery updates, and proof of delivery details. In 2023, Pos Malaysia transported over 26 million parcels and over 324 million mail pieces to over 11 million addresses across the country (Pos Malaysia Berhad, 2023). All of this data is processed using a courier management system that communicates with barcode scanners, delivery mobile apps and databases of sorting facilities to enable correct real-time monitoring of parcels throughout the delivery process.

3.3 Financial Data

One of the most comprehensive categories of data within the organisation is financial data, which is mostly managed by the SAP ERP system. These include accounts receivable and payable data, billing and invoice data, general ledger data, budget planning data, tax data, payroll data, cost centre allocations, financial reporting outputs etc. The SAP system creates about 100,000 invoices and handles payroll information for roughly 15,000 employees, based on the interview. The Finance part of SAP generates various reports, such as management reports, EA forms for tax compliance, and payroll slips—all of which are essential for both internal management and external reporting to regulatory authorities. All the financial data within the system is the basis for all statutory financial statement reporting to Bursa Malaysia and regulatory authorities.

3.4 Employee Data

Employee Data encompasses employees' details of Pos Malaysia Berhad. This entails personal details, employment agreements, job positions, payroll and benefits information, attendance and overtime data, training and development records, and performance data. This type of data is managed by the HR module of the SAP ERP system and is directly connected with the payroll processing function, enabling accurate salary disbursements to happen on time. Employee information is sensitive in particular because of its statutory deduction details for EPF, SOCSO and income tax. The HR module in SAP is vast and it caters to all the human resource activities, which are around 8000 for the HR process, based on the interview. To ensure the confidentiality of employee data, access to the system is controlled via the role-based configuration.

3.5 Operational Data

Data collected from day-to-day activities of Pos Malaysia, excluding financial and customer data. These may be system performance logs, network use data, CPU and memory use logs, input/output operations per second (IOPS) data, query response times, and system error logs. The IT team monitors the operational data actively by a set of monitoring tools to ensure the SAP ERP system and other platforms operate within the accepted thresholds as per the interview. The interview further showed that the operating environment of Pos Malaysia hosts more than 200 individual information systems and 165 system interfaces, with high volume of system level operational data generated. The information is critical for the IT department to use for system maintenance, capacity planning, and performance evaluation.

4.0 LIST OF INFORMATION SYSTEMS

4.1 List of Information Systems Used in the Organization

The interview with Pos Malaysia Berhad revealed that today they have over 200 information systems, distributed among the business units and departments. The activities include human resource management, financial management, courier and logistics, customer relationship management, and business analytics, to name a few. A large number of systems is a reflection of the complexity and scale of operations of Pos Malaysia as a national Postal and Logistics provider. Six key information systems were identified during the interviews and are displayed in the following table:

No.	Information System	Type	Brief Description
1	Enterprise Resource Planning (ERP) – SAP	Operational IS / Management IS	Core enterprise system managing HR and Finance operations.
2	Billing System	Transaction Processing System (TPS)	Automates invoice generation for postal and logistics services.
3	Courier Management System	Operational Information System (OIS)	Tracks and manages parcel and mail movement across the delivery network.
4	Customer Relationship Management (CRM) System	Management Information System (MIS)	Manages customer data, interaction records, and service histories.
5	Big Data Analytics System	Decision Support System (DSS)	Processes large data volumes to support strategic decision-making.
6	Point of Sale (POS) System	Transaction Processing System (TPS)	Handles counter transactions at post office retail branches.

4.2 Brief Explanation, Functions and Objectives of Each Information System

The various information systems of Pos Malaysia Berhad are for specific operational or management purposes. Although the systems serve different purposes, they are linked via application programming interfaces (APIs) and integration layers, enabling data to roam across these systems in near real time. Each of the systems is briefly described below, based on information obtained in the interview.

4.2.1 Enterprise Resource Planning (ERP) System – SAP

The central information system of Pos Malaysia Berhad is SAP ERP which was implemented around 2013 to centralise the HR and Finance operations at Pos Malaysia Berhad. It has a range of various functional areas such as Finance and Controlling (FI/CO), Human Capital Management (HCM), and payroll processing. The system currently has around 200 active users and controls 165 interfaces to other systems and around 11 TB of total database. Examples of key outputs are payroll slips, EA forms, general ledger reports, invoice records, and management reports for strategic decision making.

4.2.2 Billing System

Billing System is a specific system established for managing the postal and logistics services invoice generation and processing. It produces around 100,000 billing records and is completely integrated with the SAP ERP system, so that billing records are sent directly into the financial accounts. It is applicable to both corporate clients' billing and individual service billing, such as parcel delivery and logistics contracts or counter transactions.

4.2.3 Courier Management System

The Courier Management System takes care of parcel and mail tracking from start to finish. It documents shipment information at the drop-off location, monitor parcel movements within sorting centres and transit hubs and offers customers updates on delivery statuses. The system is compatible with the mobile devices that delivery staff use and connects with the CRM system to give accurate delivery information to customer service teams.

4.2.4 Customer Relationship Management (CRM) System

The Customer Relationship Management (CRM) System is a solution that integrates various tools to manage customer interactions and relationships. All customer information and records of interactions are managed by the CRM system within Pos Malaysia. It enables the customer service team to see customer profiles, make notes of customer service requests and complaints, as well as track customer service history and look for trends in customer behavior. It can be integrated with the SAP ERP and Courier Management system, which means that customer service agents have access to all information that is full and up to date when dealing with customers.

4.2.5 Big Data Analytics System

The Big Data Analytics System is designed to process massive amount of data created through the different information systems of Pos Malaysia and assist their strategic and operational decisions. It examines the trends and patterns of delivery volumes, customer behaviour, financial performance, and operational efficiency, and generates insights that can be used to guide management decisions. The SAP ERP system provides data to the Big Data system for additional analysis and visualisation based on the interview. This aligns with the company's approach to harnessing technology and data analytics through the cloud to enhance service delivery (Pos Malaysia Berhad, 2023).

4.2.6 Point of Sale (POS) System

POS system is applied at the counter shops nationwide in Pos Malaysia. It handles more than just over-the-counter sales like stamps, bill payments, parcel drop-off, insurance payments and licence renewals. Transaction records are created, which can then be put into the billing and financial systems, which mean that the revenue at the counter is captured correctly and in a standardised format.

5.0 TYPES OF INFORMATION SYSTEMS

Information systems can be categorized based on functionality and scale of operation for Pos Malaysia Berhad. It is crucial to know what type and classification of each system is, as this will help to understand how each platform contributes to the overall information architecture of the organisation. The three assigned information systems are described below: Enterprise Resource Planning (ERP) system, Billing System, and Courier Management System. The three subsections of each system (System Background, Functions and Purpose, and Classification of the System) are consistent for each system.

5.1 Enterprise Resource Planning (ERP)

5.1.1 System Background

As part of the company's initiative to streamline and to centralise the financial and human resource operations, Pos Malaysia Berhad introduced its SAP Enterprise Resource Planning (ERP) system circa 2013. SAP, an acronym for Systems, Applications, and Products in Data Processing, is a German multinational software corporation that offers ERP solutions for large businesses all over the world. SAP S/4HANA is one of the most prominent ERP platforms for large businesses, boasting a wide range of capabilities, including finance, human capital management, supply chain, and procurement (Hashmicro, 2025). Pos Malaysia's decision to implement SAP highlights the company's requirement for a comprehensive and enterprise-level system, which can handle the complexities and scale of its national operations.

An ERP system is a kind of information system that incorporates every single one of the fundamental business functions into a single, central system, as opposed to separate tools and manual techniques which give a real time visibility and control across the organisation (Selatan Technology, 2024). The SAP ERP system is precisely that function, with regards to Pos Malaysia – it is a system that connects the HR and Finance department in one integrated system. Currently, the system is used by some 200 active users, it includes 165 interfaces to other internal and external systems, and it has a total data volume of around 11 terabytes, as reported during the interview.

5.1.2 Functions and Purpose

The two main functional activities of the SAP ERP system at Pos Malaysia are HRM and Financial Management. On the HR side, the system supports employee data management, payroll processing (for around 15,000 employees), time and attendance tracking, overtime calculations, training records handling, and EA form generation for tax

submission on an annual basis. The Finance section allows to manage Accounts Receivable, general ledger maintenance, cost centre allocation, billing integration, invoice production, budget management and financial report production. Outputs of the system are payroll slips, EA forms, management reports, billing data and general ledger summaries, all of which are essential for the organisation's operational and regulatory needs.

From a system architecture perspective, the SAP ERP system relies on a Relational Database Management System (RDBMS) that is based on SQL to provide structured, consistent and query-efficient data storage. The system is hosted on both cloud and non-cloud server infrastructure and provides Pos Malaysia with some flexibility on how data is hosted and accessed. API-based integration is also available with other systems which can be automatically integrated with the system, so that the data flows between SAP and Billing System, CRM and Big Data Analytics system without manually re-entering the data across departments.

5.1.3 Classification of the ERP System

Depending on the functional layer, the SAP ERP system at Pos Malaysia can be considered as an Operational Information System and as a Management Information System (MIS). It handles the day-to-day transactions in the business like payroll calculations, invoice generation, and general ledger postings at the operational level. It creates structured reports and analytics output at the management level which can assist middle and senior management in keeping track of the performance of the department, the financial status of the department and the utilisation of human resources. It is an enterprise wide platform and it can also serve as an Executive Support System (ESS) when top level management uses the outputs of this platform for strategic planning or resource allocation.

5.2 Billing System

5.2.1 System Background

Pos Malaysia Berhad's Billing System is a customized Information System to automate and facilitate the entire billing process for postal and logistics services at the organization. Although the interview did not mention what the billing software is called, it did state that this billing system is fully integrated with the SAP ERP system so that there is no separation between the billing system and the financial accounts, thus ensuring seamless data flow. With integration, all of the invoices created using the Billing System are automatically updated in the SAP finance and accounts receivable modules, avoiding the manual reconciliation between the two systems.

5.2.2 Functions and Purpose

The main use of Billing System is to automate the process of creating an invoice for services provided by Pos Malaysia to the corporate and individual customers. The system creates about 100.000 billing records based on the interview, such as parcel delivery and courier contracts, counter transactions, and logistics services. It also handles billing for regular corporate customers, handles overdue payments and generates ageing reports for the Finance Department to keep track of overdue accounts. The automation eliminates much of the manual effort involved in the billing process, mitigating the risk of billing errors and ensuring timely and accurate revenue recognition. These outputs are subsequently transferred to the SAP ERP system, and the accounts receivable are entered in the general ledger.

5.2.3 Classification of the Billing System

Billing System is a type of Transaction Processing System (TPS). A TPS is a system used to collect, retain and process large numbers of repetitive transactions in a structured and consistent way. For Pos Malaysia, inputs are service delivery confirmations and contract billing terms, the process is billing based on predefined billing rules and the output is structured invoice documents. The system is expected to work for a long time and has a high requirement for reliability because it must not slow down or break down even when there are many transactions. This classification is in line with the definition of the TPS in information systems management literature, which emphasizes that the major use of the information system is to provide transactional accuracy, speed, and completeness, not for analytical or strategic purposes.

5.3 Courier Management System

5.3.1 System Background

The Courier Management System (CMS) is a system in operation within Pos Malaysia to oversee and track parcels and mail items through the delivery system. This is the main operational system for Pos Laju and Pos Mel courier service, which gives real-time information on the status of parcels from the acceptance counter at a post office or sorting centre to the recipient's address. This system is connected to mobile scanning devices that are used by the delivery people as well as to bar code scanning machines in the sorting centres and customer tracking interface through the Pos Malaysia website and mobile application.

5.3.2 Functions and Purpose

Major functions of the Courier Management System involve shipment creation and registration, assigning barcodes, route optimisation for the transit agents, updating the transit status, handling failed deliveries and recording the proof of delivery. The system is also connected to the CRM system, which helps customer service representatives respond to customer inquiries with delivery details in real time. The system also provides output reports like delivery status reports, operational performance dashboards, and delivery completion metrics, which are utilized by the Operations department to track and enhance service quality.

The delivery network of Pos Malaysia has transported over 26 million parcels in 2023 (Pos Malaysia Berhad, 2023). The system needs to be capable of handling this volume of deliveries, ensuring accurate data records are kept for customer tracking, financial reconciliation, and regulatory reporting needs, while also processing large amounts of movement data in real time. Forward-looking, Pos Malaysia is also still looking into implementing AI and data analytics into its logistics network to enhance the speed of parcel delivery, predictive parcel routing and issue resolution as stated in the company's Annual Report 2023 (Pos Malaysia Berhad, 2023).

5.3.3 Classification of the Courier Management System

The Courier Management System is considered to be an Operational Information System (OIS). An OIS is a kind of information system that is used to operate recurring, every day, physical processes within an organisation instead of producing analytical outputs or supporting strategic decision-making. At Pos Malaysia, the Courier Management System supports a daily operational flow for accepting parcels, tracking their status and completing their delivery in thousands of delivery routes. It guarantees that operational employees, such as those at the cashier as well as those in the field delivering items, can have access to real-time data in order to perform their jobs efficiently. The system does create a report that is used to input into management review, but its basic classification is operational since the core design of the system is to facilitate physical delivery activities, and not management analysis.

5.4 Customer Relationship Management (CRM)

5.4.1 System Background

Customer Relationship Management (CRM) is an information system that is used to unify and manage all interactions that the organisation has with its customers, contact information, communication history and service data are stored in a unified platform (Salesforce, 2023). The CRM system is one of the most important systems in use at Pos Malaysia Berhad together with SAP ERP, Billing Management, Courier Management, Big Data Analytics and POS systems. Data is provisioned to it from both SAP ERP system and the Courier Management System, enabling customer services to access real-time billing and delivery data while also handling customer enquiries without having to switch between different platforms.

5.4.2 Functions and Purpose

The CRM system at Pos Malaysia Berhad is used to provide the main system to manage all the customer interactions, service requests, their status from make to resolve, and also note the customer's complaint history in each of the customer's account. If a customer complains about a delayed or missing parcel, the system records the complaint, assigns it to the relevant team and keeps track of the situation until the problem is solved and the customer notified. In addition to complaint handling, the system also keeps a record of the manner in which the customers interact with the company and their feedback, which is then used for further analysis by the Big Data Analytics System, providing management with an insight into ongoing service issues and aiding in the decision making process for service quality improvements.

5.4.3 Classification of the CRM System

The CRM system used by Pos Malaysia Berhad is a type of Management Information System (MIS) as it is not only used to record individual transactions but it also acts as a way to collate customer information into structured reports and service performance summary that can be used to make management decisions at operational and middle management level. It does process transactions with the customers but its wider role of tracking service trends, feeding data into the analytics layer and assisting managers to review the performance of customer satisfaction also situates it more appropriately to a classification of MIS rather than a pure Transaction Processing System.

5.5 Big Data Analytics System

5.5.1 System Background

Big data analytics are the methods used to derive actionable insights from big data that are used to guide business decisions. Big data analytics includes the use of advanced platforms to process and analyse the high volume, velocity and variety of big data, which cannot be effectively processed using conventional data management tools. (IBM, 2023) In the interview, Pos Malaysia Berhad stated that its data environment is large enough to support the Big Data Analytics System, which they use and directly attributed to the size of their data environment. Data from the SAP ERP system is delivered to the system, as confirmed during the interview, to integrate data from all platforms throughout the organisation into one analytical environment.

5.5.2 Functions and Purpose

The Big Data Analytics System at Pos Malaysia integrates data from various source systems, such as the SAP ERP, Courier Management System and CRM system, to uncover patterns in operations, money and customer behaviour that wouldn't be seen at the transaction level. It, for instance, analyses the performance of delivery routes and enables data-driven decisions to be made in the Operations department based on this analysis; it also processes financial data from SAP and generates cost analysis and revenue trend summaries for the Finance department. This system is also a major element within the company's digital transformation strategy, known as POS 2.0, which is based on the use of data to enhance the customer experience, delivery efficiency and overall business sustainability (Pos Malaysia Berhad, 2023).

5.5.3 Classification of the Big Data Analytics System

The Big Data Analytics System at Pos Malaysia Berhad belongs to one of the types of information systems, namely Decision Support System (DSS), which is an information system that uses data analytical models and tools for supporting the manager's decision-making process in complex situations or uncertainty. This classification is suitable because the system is not designed to process mass volumes of routine transactions or create standard operational records, but rather to analyse huge amounts of information scattered across multiple systems, and present this information to the management that will make decisions about how to change their delivery processes, allocates operational budgets, or invests in new service capabilities.

5.6 Point of Sale (POS) System

5.6.1 System Background

A Point of Sale (POS) system is a tool that facilitates the processing and the real-time recording of the transactions of a customer at the point of service, serving as the link between customer and organisation's financial systems (Oracle, 2023). The network of more than 3,500 service touchpoints nationwide of Pos Malaysia Berhad includes post office branches, postal agents and Self-Service Outlets (Pos Malaysia Berhad, 2023). The Postal segment is vital because it accounted for RM1,086.11 million or 58.1% of the total group revenue in 2023 and is important to ensure that the high volume of counter-level revenue is captured with accuracy and then used in the overall financial management system.

5.6.2 Functions and Purpose

The POS system at Pos Malaysia can be used to transact various of the OTC activities such as the purchase of stamps, packaging, acceptance of parcels for both domestic and international delivery, payment of bills, renewing road tax, paying insurance premiums, and other retail activities provided at the post office branches. All transactions are captured in real time, a receipt is issued to the customer, and the transaction data is automatically sent to the Billing System and SAP ERP system for financial record and reconciliation. The system also produces a daily transaction summary at each counter, which helps the counter managers in reconciling daily collections, and in reporting the relevant revenue amounts to the Finance department at the end of each business day.

5.6.3 Classification of the POS System

The POS system used by Pos Malaysia Berhad is considered as Transaction Processing System (TPS) because it mostly deals with the real-time capture and processing of high volumes of repetitive and structured customer transactions, and carries out these processes, thousands of times per day across thousands of counter transactions. It doesn't perform analytical functions or create management reports, but rather it makes sure that every transaction is recorded properly, receipts are issued accurately and that data is passed on to the financial systems for further processing, which is define as a TPS being a system that is used at the operational level of the organisation to process the routine transactions needed for business.

6.0 DETAILS OF SAP ENTERPRISE RESOURCE PLANNING (ERP)

6.1 Introduction to SAP ERP

Pos Malaysia Berhad implemented SAP Enterprise Resource Planning (ERP) as the foundational information technology to handle its essential business processes. SAP is short for Systems, Applications and Products in Data Processing. It is a world famous business software solution developed by a German multinational company with the same name. The system was introduced by Pos Malaysia in about 2013 as part of the company's efforts to centralise and simplify its Human Resource (HR) and Finance processes into a single integrated platform. In a national organisation such as Pos Malaysia, these functions before the integrated ERP system would have been handled with scattered systems and much manual processing, which would have increased the chances of errors and decreased the efficiency of decision-making.

According to the interview with the Pos Malaysia IT professional, the SAP ERP is designed to replace individual, department specific tools with a single system that provides a unified view of organisational data in real time. Instead of Human Resources having its own payroll database and Finance having its own billing records, SAP brings HR and Finance together meaning the data may flow easily between the departments. For Pos Malaysia, this integration is particularly important given the size of the workforce, which numbers in the thousands, and the volume of financial transactions the company processes each month. The platform processes the payroll for around 15,000 employees and creates about 100,000 invoices, showing the volume transactional needs of the platform.

Furthermore, it is integrated with other information systems inside Pos Malaysia such as the application programming interface (APIs) like the Billing System, the Customer Relationship Management (CRM) system and the Big Data Analytics. The connectivity allows data created in SAP to be made available to other systems without the need for manual re-entry, which improves both efficiency and data accuracy throughout the organisation. The SAP ERP system is the technology backbone of Pos Malaysia's HR and Finance operations and provides the foundation to meet the complex and high-volume business needs of a national-scale company.

6.2 Functional Description of the Information System

6.2.1 Human Resource Management

The Human Resource Management function in the SAP ERP system is very crucial in handling employee information and daily HR operations at Pos Malaysia. The business manages the entire employee lifecycle, from recruitment to retirement, using the Human Capital Management (HCM) module. Having all HR data in one integrated system will enable Pos Malaysia to achieve better accuracy, eliminate redundant data and make HR processes more efficient across the company.

The HR department generates the employee master record in the HCM module when a new employee joins the organization. This record has vital information which includes personal information, job post, department, employment terms and salary grade. These records are the main source of information for other HR actions. During the interview there was no indication of an automated recruitment process. SAP provides a systematic procedure for onboarding that captures and structures employee information to help ensure that it is complete and consistent.

Moreover, The HCM module also assists with staff development by capturing training and certification information. According to the interview, the employees need not only technical IT expertise but also the knowledge connected to their job duties such as accounting abilities for the users of the Finance module. These records can be used by HR for tracking completed training, identifying skill gaps and planning future learning programmes. This is quite significant, especially considering there are about 200 people that are actively using the SAP system, and each module requires different knowledge and responsibility.

Lastly, Payroll administration is among the most crucial aspect of HCM module. The system processes payroll for almost 15,000 employees with automatic computation of gross pay, overtime and regulatory deductions such as EPF, SOCSO and Monthly Tax Deduction (MTD/PCB). Payroll is immediately connected with attendance, allowed leave and overtime, which reduces manual effort and the danger of calculation mistakes. It also covers the employee benefits like medical claims, housing allowances etc. Approved payments are part of the payroll cycle. SAP will generate Borang EA for all employees at the end of each year based on the information maintained in the system. This supports Pos Malaysia to produce tax documents fast and accurately, with the HCM module being important in boosting efficiency and meeting regulatory standards.

6.2.2 Financial Management

Pos Malaysia also uses SAP ERP system which has another vital role which is financial management. This function is maintained by the Finance and Controlling (FI/CO) module and helps the company to process financial transactions, monitor business performance and fulfil regulatory requirements. When all financial data reside in an integrated system, the Finance department can operate more efficiently with information and to a higher degree of accuracy and consistency across all company processes.

Additionally, Billing and Creating the invoice is one of the major actions of FI/CO module. The interview produces around 100,000 billable records per processing cycle for services given to both individual and commercial customers. These billing records are system generated from service delivery data, which means less manual labour and reduced potential for human error. SAP checks the billing information and creates official invoices and posts them to accounts receivable. This way each invoice is attached to the right client account and can be followed through until the money is paid.

The module is also a critical component in controlling corporate costs and managing accounts receivable. The Finance team tracks outstanding customer payments using SAP, records incoming payments and generates ageing reports to determine overdue invoices. This information helps the organisation's cash flow as it allows it to follow up on late payments more effectively. At the same time, the Controlling (CO) module allows each department to post costs to the right cost center. When actual expenditure is compared with the approved budget, managers are able to more easily identify unnecessary costs and take corrective action before financial problems worsen.

Another important role of the FI/CO module is to provide tax management and financial reporting. The system computes statutory tax liabilities and logs the liabilities along with the relevant financial transactions. This helps Pos Malaysia to meet legal and regulatory requirements. It also creates accurate financial reports which are used by the senior management to evaluate the business performance and make informed decisions. The company processes financial information on a single integrated platform, which helps it to minimise reporting errors, improve financial management and meet the governance requirements expected of a major public listed organization.

6.2.3 System Integration

System integration is one of the most important aspects of SAP ERP system at Pos Malaysia whereby different business systems are able to work as one. SAP combines these systems to ensure users do not need data in many programs and are utilising separate apps. Data can be sent easily from department to department. This prevents duplication of labour, improves data accuracy and helps ensure employees receive the same information all the time. This allows the organization to run more efficiently and make better decisions based on reliable data.

According to the interview identified 165 active linkages between the SAP ERP system and internal and external services. This shows the importance of SAP to the entire information architecture of Pos Malaysia. Many of these links are via Application Programming Interfaces (APIs) that enable data to be transmitted across the different software applications in a secure and structured fashion. For example, Billing System feeds billing information directly into SAP so that bills are posted instantly into accounts receivable module. The client information is also provided to SAP from the client Relationship Management (CRM) system to correctly handle accounts and billing. These automated data transfers eliminate the need for manual data entry and limit the possibility of errors.

Besides that, the system also interacts with the Big Data Analytics platform. SAP takes data and transmits it to the analytics platform where it may be explored in further depth in order to assist with company planning and performance evaluation. Besides API interaction, Pos Malaysia also uses File movement Protocol (FTP) for the movement of huge amounts of data between the systems. This is great for operations like this that do not require real-time changes. For example, sending daily recaps of transactions from branch offices to the main SAP database. The organization can use APIs and FTP to deal with different kinds of data effectively, while maintaining the completeness and consistency of all information.

6.2.4 Report Generation

One of the most significant benefits of the SAP ERP system is its ability to generate reports. The system can convert daily business transactions into useful reports for management and regulatory needs. SAP do not require you to go to different departments to get data manually. SAP processes the data automatically and provides accurate and straightforward results. This saves time and reduces human mistake and provides managers with reliable information for decision making. This makes the reporting process more efficient and speedier across the organisation.

Furthermore, The Finance module generates a variety of reports that management utilises to monitor the financial performance of the organization. This includes financial summaries, cost centre reports, accounts receivable reports and general ledger reports. Most reports are created every month or quarter, to present a picture of the financial status of the organization. The reports enable managers to compare actual spending to budgeted spending, spot financial trends, and monitor overdue client payments. Another useful feature of SAP is the ability to break down from the summary reports to the individual transactions. It allows managers to investigate unusual data, understand why variations from the budget exist and make better informed company decisions based on comprehensive financial information.

Last but not least, the Human Resource department also has crucial information like the staff yearly Borang EA. SAP also added that the report is published for around 15,000 personnel at the end of the financial year. The documentation will provide a summary of the employee's annual salary, allowances and statutory deductions to be completed for their income tax filing with LHDN accordingly. The data is extracted directly from the SAP payroll records and correlates with an organization's financial records. This makes the tax requirements less complex and reduces the risk of errors in reporting. Besides creating Borang EA, the system also produces payroll summary reports and invoices for HR and Finance departments to audit salary payment and billing process.

7.0 INFORMATION SYSTEM COMPONENTS

7.1 Hardware

7.1.1 Server Infrastructure

Pos Malaysia's SAP ERP system is run on a server infrastructure separated into production and non-production environments. The production servers run the live system, where all real business operations take place. This comprises daily transactions such as payroll processing of over 15,000 employees and generation of around 100,000 invoices in each processing cycle. The workload makes that the production environment should be stable and responsive at all times. A brief system outage could jeopardise critical operations and business continuity.

In contrast, non-production servers are used for development and testing purposes. These environments allow the IT team to try out system changes, apply configuration updates, and test new features before they are moved into the live system. By doing this, any errors or performance issues can be identified earlier without affecting actual business operations. This separation between testing and live environments is a common practice in enterprise systems because it helps reduce operational risk and maintain system reliability.

Based on the interview, the IT department at Pos Malaysia manages both environments closely. Any changes to the system are first reviewed and tested in the non-production setup before being approved for deployment into the production environment. This controlled approach ensures that updates do not introduce unexpected issues into the live system. It also helps maintain consistent system performance and reduces the likelihood of disruptions during critical business processes.

7.1.2 Cloud and Non-Cloud Environment

The on-premises infrastructure is mainly used for systems and information that require a higher level of control. Since employee records, payroll information, and financial data are highly confidential, storing some of this information on physical servers allows the organisation to manage security and system access more directly. At the same time, cloud services provide additional flexibility because computing resources can be increased whenever demand becomes higher. This means the organisation does not have to depend entirely on physical hardware, which has limited capacity and usually requires higher maintenance costs.

The interview findings also match Pos Malaysia's digital transformation strategy. In 2022, the organisation expanded its technology infrastructure by adopting Amazon Web Services (AWS) as part of its cloud migration initiative. Rather than replacing the existing infrastructure completely, Pos Malaysia continues to operate in a hybrid environment where workloads are

shared between cloud services and on-premises servers. This approach allows the company to improve system performance while keeping infrastructure costs under control and maintaining proper management of organisational data.

The hybrid environment is particularly useful during periods when system activity increases significantly. Large processes such as monthly billing and payroll processing require considerable computing power to ensure that transactions are completed without delays. By using both cloud and non-cloud infrastructure, the SAP ERP system can maintain stable performance even during busy periods.

7.2 Software

7.2.1 SAP ERP

The main software used in this information system is SAP ERP, an enterprise resource planning system developed by SAP SE. Pos Malaysia adopted this system to support its HR and Finance operations because it can manage large volumes of business data and transactions within a single platform. As a national postal and logistics provider, the organisation requires software that can support different departments while maintaining accurate and consistent information.

According to the interview, SAP ERP has been used at Pos Malaysia since around 2013. Since its implementation, the system has been updated and customised to meet changing business needs and regulatory requirements. The IT team is responsible for maintaining the software, applying updates, and configuring the system whenever new business requirements arise. This ensures that the system continues to support the organisation effectively as its operations evolve.

7.2.2 SAP Modules

Pos Malaysia mainly uses two SAP modules: Human Capital Management (HCM) and Finance and Controlling (FI/CO). The HCM module supports HR activities such as employee records, payroll processing, attendance, benefits administration, and the generation of Borang EA. It is also configured to calculate statutory deductions such as EPF, SOCSO, and Monthly Tax Deduction (MTD/PCB) based on Malaysian regulations.

The FI/CO module manages financial activities, including billing, accounts receivable, general ledger, cost centre management, and financial reporting. The interview indicated that users of this module require accounting knowledge together with SAP skills because many financial processes involve specialised functions. An important advantage of these two modules is their integration. For example, payroll information processed in HCM is transferred automatically to

the Finance module without requiring manual data entry, improving both accuracy and efficiency.

7.2.3 SAP GUI

Employees access the SAP ERP system through the SAP Graphical User Interface (SAP GUI), which serves as the main desktop application for daily operations. Through this interface, users can enter data, process transactions, and generate reports based on their assigned roles. The interview revealed that approximately 200 active users access the SAP system using SAP GUI.

Although SAP GUI provides many useful features, it can be challenging for new users because each module contains different transaction codes and navigation paths. As a result, employees need proper training before they can use the system confidently. Regular training also helps users work more efficiently and reduces the likelihood of processing errors during daily operations.

7.2.4 System Integration via API

Besides the SAP software itself, Pos Malaysia uses Application Programming Interfaces (APIs) to connect SAP with other business systems. APIs allow different applications to exchange information automatically without requiring employees to enter the same data multiple times. This improves data accuracy and speeds up business processes across the organisation.

According to the interview, SAP is connected to systems such as the Billing System, Customer Relationship Management (CRM) system, and the Big Data Analytics platform through API integration. For instance, billing information generated by the Billing System is transferred directly into SAP's accounts receivable module. The interview also mentioned that Pos Malaysia currently manages 165 active system interfaces, showing that API integration plays an important role in ensuring information flows smoothly between different business applications.

7.3 Database

7.3.1 Database Platform

The SAP ERP system at Pos Malaysia stores its information in a relational database management system (RDBMS). This type of database organises information into related tables, making it easier to store, retrieve, and update data accurately. By linking different tables together, the database can maintain consistent information across multiple business functions while reducing duplicate records. According to the interview, the SAP ERP system uses a SQL-based database, which is commonly used in enterprise systems because it supports reliable data management and high processing performance.

Structured Query Language (SQL) is the standard language used to communicate with the database. It allows the system to add new records, update existing information, search for specific data, and remove records when necessary. Since the SAP ERP system stores around 11 terabytes of organisational data, SQL plays an important role in ensuring that large amounts of information can be processed efficiently. This allows employees to access accurate data quickly while supporting the smooth operation of HR, Finance, and other business functions.

7.3.2 Types of Data Stored

The SAP database stores different types of information that support the daily operations of Pos Malaysia. Human resource data includes employee profiles, employment details, payroll records, attendance, leave applications, overtime information, training records, and employee benefits. Because this information contains confidential personal data, appropriate security measures and access controls are necessary to ensure that only authorised personnel can view or update the records.

The database also stores financial information generated through the Finance module. This includes general ledger records, invoices, accounts receivable, payroll payments, cost centre information, and financial reports. Based on the interview, the system processes approximately 100,000 invoices and payroll information for around 15,000 employees, resulting in a large amount of financial data that must be stored accurately for auditing and reporting purposes.

Besides HR and financial information, the database also contains customer and operational data. Customer records mainly consist of billing information, payment histories, and account details that support invoicing and payment collection. Operational data includes system logs, interface records, and performance information that help the IT team monitor the SAP environment.

7.5 Network

7.4.1 Local Area Network (LAN)

The Local Area Network (LAN) forms the core of internal connectivity within Pos Malaysia's offices and data centre. It connects computers, servers, and other devices within a limited physical area such as a building or office complex. Through this network, SAP users are able to access the system's servers and perform their daily tasks with stable and fast communication between devices. Since most SAP processing happens through this internal connection, the quality of the LAN has a direct impact on how smoothly users experience the system.

In the context of Pos Malaysia, the LAN supports operations in both the data centre and administrative offices. It ensures that HR and Finance users can access SAP without delays when entering data, processing transactions, or generating reports. A stable LAN connection also reduces interruptions during system usage, which is important for maintaining productivity in daily operations. In general, the LAN acts as the main internal communication layer that supports efficient access to the SAP ERP system.

7.4.2 Wide Area Network (WAN)

The Wide Area Network (WAN) is used to connect Pos Malaysia's various branches, post offices, and remote locations across the country. Unlike the LAN, which is limited to a single site, the WAN allows communication over long distances. This is important for an organisation that operates nationwide and serves a large customer base across Malaysia.

Through the WAN, remote locations can still access centrally managed systems such as SAP ERP when needed. Although most SAP processing is handled at the headquarters, branch-level systems and users still rely on WAN connectivity to exchange data with the central system. The performance of the WAN therefore affects how quickly information can be transmitted between locations, especially during peak operational periods. A stable and well-managed WAN helps ensure that all branches remain connected to the same central system without major delays.

7.4.3 API Integration and FTP Communication

In addition to LAN and WAN connectivity, SAP ERP at Pos Malaysia also relies on network-based communication through APIs and File Transfer Protocol (FTP). These methods are used to exchange data between SAP and other internal or external systems. Based on the interview, the system currently operates with around 165 active interfaces, many of which depend on these communication methods.

FTP is commonly used for transferring large sets of data between systems. This is usually done in batches, such as sending payroll files or receiving billing data from external systems. It is suitable for non-real-time processes where immediate updates are not required. On the other hand, API communication supports more direct and real-time data exchange between systems. This allows information from applications such as the Billing System and CRM to be updated automatically in SAP without manual input. Both communication methods depend heavily on stable network infrastructure. Any disruption in network connectivity can affect data transfer between systems and lead to delays in processing. Because of this, the network layer plays an important role not only for end users but also for ensuring smooth integration between SAP and other systems within Pos Malaysia's IT environment.

7.6 People

7.5.1 IT Team

The SAP ERP system at Pos Malaysia is managed by a small IT team consisting of around eight members, based on the interview findings. This team is responsible for maintaining the system's daily technical operations, including monitoring servers, handling configuration tasks, and ensuring overall system performance remains stable. They also coordinate with external SAP vendors when specialised technical assistance is required.

Despite the relatively small team size, they support a system that handles large-scale organisational processes. This is possible because many monitoring and maintenance tasks are partially automated, while more complex issues are escalated to external support. As a result, the IT team focuses mainly on system stability, operational continuity, and coordination with other technical stakeholders.

7.5.2 System Administrators

Within the IT team, system administrators play a key role in managing the SAP environment. Their responsibilities include controlling user access, maintaining system security, applying updates, and monitoring both server and database performance. They also handle incidents when technical problems occur and ensure that system issues are resolved within a reasonable timeframe.

A major part of their role involves managing transport requests, which are used to move system changes from the testing environment to the live production system. This process requires careful control because incorrect configuration changes can affect business operations. Therefore, system administrators must have a strong understanding of both the SAP technical structure and the organisation's business processes.

7.5.3 Vendors and Contractors

Due to the complexity of SAP ERP, Pos Malaysia also depends on external vendors and contractors to support its internal IT team. These external experts provide assistance in areas such as system configuration, licensing support, performance tuning, and troubleshooting more complex technical issues. Based on the interview, the SAP support structure includes both internal staff and external specialists. This combination is commonly used in enterprise systems because it allows organisations to maintain internal control while still accessing advanced technical expertise when needed. It also ensures that system issues can be addressed more effectively without overloading the small internal team.

7.5.4 End Users

End users of the SAP ERP system at Pos Malaysia are mainly HR and Finance employees who use the system through the SAP Graphical User Interface (SAP GUI). This group includes payroll staff, HR officers, finance executives, and accounts receivable personnel. Each user interacts with specific modules and transaction codes based on their job responsibilities. Although end users do not deal with the technical infrastructure of the system, their daily operations depend heavily on system availability and performance. They rely on SAP to process transactions, manage records, and generate reports accurately. Any system delay or error can directly affect their workflow, especially during high-volume processing periods.

7.5.5 Skills Required

One of the key points highlighted in the interview is that SAP ERP requires a specialised combination of technical and domain knowledge. Unlike basic software applications, SAP users must understand both the system functions and the business rules behind the processes they handle. For example, Finance users need knowledge of accounting principles as well as SAP financial configuration, while HR users must understand payroll rules, statutory deductions, and HCM module settings. The interview also noted that Finance module users are expected to have accounting backgrounds combined with IT-related skills, and in some cases, formal SAP certification is required.

This requirement increases the importance of training and development within the organisation. New employees cannot use the system effectively without proper preparation, and staff turnover in SAP-related roles may affect operational continuity. Because of this, ongoing training and certification are important to ensure that users remain competent and that the system continues to be used correctly across all departments.

7.7 Procedures

7.6.1 Daily Operations

Daily operations in the SAP ERP system at Pos Malaysia consist of routine processing activities that support core business functions. These include payroll runs, invoice generation, updates to accounts receivable, attendance processing, and scheduled report execution. Each process follows a fixed timetable within the system to ensure consistency and timely completion of tasks.

These activities are carried out through a combination of automated system workflows and user involvement. End users perform data entry and verification, while the system processes the information according to predefined rules. The IT team oversees the overall operation to ensure that each process runs smoothly and completes without interruption. This structured approach helps maintain operational stability across HR and Finance functions.

7.6.2 System Monitoring

System monitoring is performed continuously by the IT team to ensure that SAP operates within normal performance levels. Based on the interview, key performance indicators include CPU usage, memory consumption, I/O performance (IOPS), and system response time. These metrics are tracked using monitoring tools that provide real-time updates on system health.

When abnormal readings are detected, alerts are triggered so the IT team can investigate the issue immediately. This helps prevent performance degradation or system downtime. In addition to performance monitoring, the team also reviews system logs that record errors, failed transactions, and interface issues. Given that SAP is connected to around 165 system interfaces, monitoring integration errors is especially important because a single failure can affect data flow between multiple systems.

7.6.3 Configuration Management

Configuration management refers to the controlled process of modifying system settings and business rules within SAP. At Pos Malaysia, configuration changes are carried out regularly to reflect updates in operational requirements, such as payroll rules, billing structures, or reporting formats.

All changes are first implemented in a non-production environment where they are tested before being moved into the live system. This testing stage is important because it helps identify potential issues early and reduces the risk of disrupting business operations. Only after approval are the changes transported into the production environment.

7.6.4 Maintenance and Updates

System maintenance is carried out regularly to ensure that SAP ERP remains stable, secure, and up to date. These activities include applying vendor-provided patches, fixing known issues, and implementing system improvements. The IT team is responsible for scheduling and executing these updates as part of their ongoing duties.

Maintenance work is usually planned during periods of low system usage to minimise disruption to users. In some cases, the system may need to be temporarily taken offline. Careful planning is therefore required to ensure that essential business operations are not significantly affected.

7.6.5 Backup

Data backup is an essential procedure that protects the organisation against data loss. Pos Malaysia maintains regular backups of the SAP database based on a defined schedule. These backups act as recovery points that can be used to restore the system if data corruption, hardware failure, or accidental deletion occurs.

Because the system stores a large volume of data, estimated at around 11 terabytes, backup management requires sufficient storage capacity and careful coordination. This ensures that backup files remain reliable and can be restored when needed without data inconsistency.

7.6.6 High Availability (HA)

High Availability (HA) is implemented to reduce system downtime and ensure continuous access to SAP ERP. The system is designed with redundant components so that if one part fails, another can take over automatically. This failover mechanism helps maintain system availability even during unexpected technical issues.

For Pos Malaysia, HA is important because critical processes such as payroll and billing depend on uninterrupted system access. With HA in place, users experience minimal disruption, and business operations can continue even if hardware or system components encounter failure.

7.6.7 Disaster Recovery

Disaster Recovery (DR) refers to the procedures used to restore the SAP ERP system after a major disruption. This may include events such as data centre failure, severe system damage, or large-scale data loss. According to the interview, Pos Malaysia has a DR plan that outlines how the system should be recovered in such situations.

The DR plan includes recovery time objectives (RTO) and recovery point objectives (RPO), which define how quickly the system must be restored and how much data loss is acceptable. These guidelines help ensure that recovery efforts are structured and aligned with business needs. Overall, the disaster recovery framework is an important part of risk management, ensuring that SAP ERP can be restored and operational within an acceptable timeframe.

8.0 INFORMATION SYSTEM ACTIVITIES

8.1.1 What is Input?

Input refers to the raw data that is entered into an information system before any processing takes place. It forms the starting point of the information processing cycle, and its accuracy directly influences the reliability of the system's output. If the input data is incomplete or incorrect, the results produced by the system will also be unreliable.

In the SAP ERP system at Pos Malaysia, input includes all types of data that enter the system from different sources. This may involve manual entry by users, automated data from integrated systems, or batch transfers from external platforms. Because SAP relies heavily on structured data processing, the consistency and correctness of input data play an important role in ensuring smooth system operations and accurate reporting.

8.1.2 Manual Input

Manual input refers to data that is entered directly into SAP by users through the SAP Graphical User Interface (SAP GUI). This is commonly used in HR activities where officers input employee details such as attendance records, leave applications, overtime approvals, and new staff information. In the Finance module, manual input includes journal entries, budget adjustments, and cost allocations that are not generated automatically by the system.

Based on the interview, HR and Finance staff use SAP on a daily basis to enter and update this information. These inputs are essential because they capture real-world decisions and activities that occur outside the system's automated processes. However, manual entry also increases the risk of human error. To reduce this risk, SAP includes validation checks that review data before it is accepted into the system, helping to detect mistakes early and maintain data accuracy.

8.1.3 API Input

API input refers to data that is transferred automatically into SAP from other systems through Application Programming Interfaces. This is one of the most important input methods in Pos Malaysia's SAP environment because the system is connected to 165 active interfaces.

Through these connections, data flows continuously between SAP and other applications without requiring manual intervention.

For example, when the Billing System generates an invoice, the information is automatically sent to SAP's accounts receivable module. Similarly, customer data from the CRM system and operational data from other platforms can be updated directly in SAP. This automated process ensures that the information in SAP remains up to date and consistent across systems, which is especially important in a large organisation handling high transaction volumes.

8.1.4 Other Integrated Systems

In addition to API-based input, SAP also receives data through batch processing and file-based transfers using FTP. This method is used when large volumes of data need to be transferred at scheduled intervals rather than in real time. For instance, daily or end-of-day summaries from branch systems may be sent to the central SAP database in a single file transfer.

According to the interview, Pos Malaysia operates more than 200 information systems, many of which contribute data to SAP through these integration methods. This combination of real-time and batch input allows the organisation to manage different types of data efficiently based on their urgency and processing requirements.

8.1.5 Data Validation

Data validation is a key control mechanism during the input stage that ensures all incoming data meets required standards before it is processed by the system. SAP uses built-in validation rules to check whether data is complete, correctly formatted, and logically consistent. For example, it verifies employee ID formats, checks valid date ranges, and ensures financial values fall within approved limits.

If the input data does not meet these requirements, the system will reject it and prompt correction before it can be processed further. This is important in the context of Pos Malaysia because inaccurate data can lead to serious operational issues, such as payroll errors or incorrect financial reporting. By filtering errors at the entry stage, data validation helps maintain the integrity of the entire system and reduces the risk of costly mistakes in later processes.

8.2 Processing

8.2.1 What is Processing?

Processing refers to the set of operations carried out by an information system to convert raw input data into meaningful information. In SAP ERP at Pos Malaysia, this stage involves calculations, data manipulation, and logical rules applied after data has been entered into the system. It represents the core function of the system where information is transformed into usable outputs that support business decisions and operations.

Within SAP, processing is continuous and highly automated. Once data is received from users or other systems, the ERP engine processes it according to predefined rules embedded in the system. This allows large volumes of data to be handled efficiently without requiring manual computation, which is especially important for an organisation that manages nationwide operations and high transaction volumes.

8.2.2 HR Processing

HR processing in the SAP HCM module focuses on converting employee-related data into structured HR outputs. Attendance records are processed to determine working hours, overtime, and leave balances for each employee. These results are then used as input for payroll calculations. Similarly, approved benefit claims are processed to update employee entitlements and ensure that payments are reflected correctly in the system.

Training records are also updated through processing activities when employees complete courses or certifications. At the end of the financial year, the system compiles payroll and deduction data to generate EA form information for each employee. According to the interview, SAP handles around 8,000 HR-related processes, showing the scale of HR operations managed through the system. This level of automation reduces reliance on manual calculations and helps improve accuracy in HR administration.

8.2.3 Financial Processing

Financial processing within the FI/CO module involves converting transactional data into structured accounting records. When payroll runs are completed or billing data is received from external systems, the information is automatically posted to the general ledger. This ensures that financial records are updated in real time and remain consistent across the organisation.

Cost centre accounting is another key function, where expenses are assigned to specific departments based on predefined rules. This allows management to monitor spending and compare actual costs against budgets. In accounts receivable, the system processes

incoming payments, matches them with outstanding invoices, and updates customer balances automatically. These processes ensure that financial information remains accurate and aligned with standard accounting practices.

8.2.4 Payroll Processing

Payroll processing is one of the most critical functions in SAP ERP at Pos Malaysia. The process begins with attendance data, overtime records, and approved claims, which are used to calculate gross salaries for approximately 15,000 employees. The system then applies statutory deductions such as EPF, SOCSO, and income tax based on LHDN requirements to determine net pay.

After calculations are completed, the system generates payslips and payroll reports, and the total payroll cost is posted to the Finance module. Accuracy is essential in this process because any errors directly affect employee payments and compliance obligations. Automation in SAP reduces the likelihood of miscalculations and ensures that payroll is processed consistently and efficiently.

8.2.5 Billing Processing

Billing processing involves the creation and management of invoices for services provided by Pos Malaysia. Data received from the Billing System is processed within SAP to generate official invoices, update customer accounts, and record revenue in the financial system. Based on the interview, the system processes around 100,000 billing records per cycle, highlighting the importance of automated processing for handling large transaction volumes.

Pricing rules and discount conditions are also applied automatically during processing. These rules ensure that all invoices follow the correct contract terms and remain consistent across customers. Once processing is completed, the system produces accurate invoice documents that are ready for distribution and financial reporting.

8.2.6 Tax Processing

Tax processing in SAP ERP covers both payroll-related and financial transaction taxes. For payroll, the system calculates monthly income tax deductions based on employee earnings and applies LHDN's MTD/PCB requirements. This ensures that tax deductions are accurate and compliant with national regulations.

For financial transactions, the system records tax amounts linked to invoices and other financial activities, supporting corporate tax reporting. At the end of the year, SAP compiles payroll data to generate EA form information for all employees, summarising their total income

and deductions. This automated process reduces manual workload and improves accuracy in tax reporting for both employees and the organisation.

8.2.7 Report Processing

Report processing refers to how SAP compiles and generates reports based on stored data in the system. The system can produce reports either on demand or according to scheduled intervals. During this process, transaction-level data is grouped, filtered, and summarised into structured formats that are easier to analyse.

At Pos Malaysia, report processing supports HR and Finance reporting needs, including management summaries, financial statements, and operational reports. These reports are used for decision-making, performance monitoring, and regulatory compliance. By automating report generation, SAP ensures that accurate and up-to-date information is always available when required.

8.3 Output

8.3.1 What is Output?

Output refers to the final results produced by an information system after data has been processed. It represents the end stage of the information cycle, where processed data is converted into meaningful information that can be used for decision-making, reporting, and operational purposes. Outputs can take different forms depending on the system and the type of processing involved.

In the SAP ERP system at Pos Malaysia, outputs include a wide range of documents and reports generated from HR and Finance activities. These outputs are used internally by staff and management, as well as externally for customers and regulatory bodies. Common examples include invoices, payslips, EA forms, and management reports, all of which are automatically generated based on processed system data.

8.3.2 Invoice Generation

Invoice generation is one of the most important and high-volume outputs produced by the SAP ERP system. Based on the interview, the system generates around 100,000 invoices in each processing cycle, covering a wide range of services provided to both individual and corporate customers. Each invoice contains details such as service description, amount payable, due date, and customer account information.

These invoices are created automatically using data from the billing process in the accounts receivable module. Standard templates within SAP ensure that all invoices follow a consistent

format. This output is essential because it communicates payment obligations to customers and forms the basis for revenue recognition in the financial system. The efficiency of this process is crucial for maintaining cash flow and ensuring accurate financial records.

8.3.3 Payroll Results

Payroll results are generated after each payroll cycle is completed. These outputs include individual payslips for approximately 15,000 employees, showing gross salary, deductions, and net pay. In addition to payslips, the system also produces summary reports for HR and Finance departments, which show total payroll costs, statutory deductions, and breakdowns by department or cost centre.

These outputs are important for both internal and external purposes. Internally, they support salary payments and financial reporting, while externally, they ensure compliance with statutory bodies such as EPF, SOCSO, and LHDN. Because payroll data directly affects employees, accuracy is essential. SAP helps ensure this by automatically calculating and validating all payroll-related figures.

8.3.4 EA Forms

EA forms, also known as Borang EA, are statutory documents that must be issued to employees at the end of each financial year. These forms summarise an employee's total annual income, allowances, and statutory deductions such as EPF, SOCSO, and income tax contributions. Employees use these forms when submitting their personal tax returns to the Inland Revenue Board of Malaysia (LHDN).

In SAP, EA forms are generated automatically from stored payroll data. The system compiles information collected throughout the year and formats it according to regulatory requirements. For Pos Malaysia, this automated process is particularly important due to the large number of employees. It reduces manual workload while ensuring that all forms are consistent, accurate, and compliant with tax regulations.

8.3.5 Management Reports

Management reports are structured outputs designed to support decision-making at different levels of the organisation. These reports are generated from processed HR and Finance data and can be produced either on demand or according to a schedule. They provide a clear overview of organisational performance and help management monitor key indicators.

Typical reports include financial summaries, cost centre performance reports, accounts receivable ageing reports, and HR analytics such as workforce size, overtime trends, and

leave usage. These reports present large volumes of data in a simplified format, making it easier for managers to interpret and act on the information.

Without SAP, producing these reports would require manual data collection from multiple systems, which would be time-consuming and less accurate. The automated reporting function therefore improves both the speed and reliability of decision-making at Pos Malaysia.

8.4 Storage

8.4.1 What is Storage?

Storage refers to the process of retaining data within an information system after it has been processed. It ensures that information remains available for future use, whether for reference, reporting, or additional processing. Without proper storage, the outputs generated by a system would be temporary and could not support long-term decision-making or compliance requirements.

In the SAP ERP system at Pos Malaysia, storage includes both active storage of current transaction data and long-term retention of historical records. This is important because the organisation needs to preserve financial, HR, and operational information not only for daily operations but also for audit purposes and regulatory compliance. As a result, storage plays a continuous role in supporting the overall information lifecycle.

8.4.2 SQL Database Storage

The main storage component of the SAP ERP system is a SQL-based relational database. All data entered into the system, whether through manual entry, API integration, or batch uploads, is stored in a structured format within this database. This allows the system to organise and retrieve information efficiently when needed.

According to the interview, the database currently stores around 11 terabytes of data. This includes HR records, financial transactions, customer billing information, and operational logs collected since the system was implemented in 2013. The large volume of stored data reflects years of continuous business activity and highlights the importance of having a stable and scalable storage system.

8.4.3 RDBMS Structure

The SAP database is built on a Relational Database Management System (RDBMS), where data is organised into tables that represent different business entities. For example, separate tables may store employee records, invoices, cost centres, and financial accounts. These tables are connected through relationships using primary and foreign keys.

This structure allows the system to link related data across different modules. For instance, payroll data from the HR module can be connected with cost centre information from the Finance module. This relational design supports complex queries and reporting functions, enabling SAP to generate integrated reports that combine data from multiple sources within the organisation.

8.4.4 Data Security

Data security within the SAP storage system is maintained through a combination of technical and administrative controls. One of the main mechanisms is role-based access control, which restricts user access based on job responsibilities. This means users can only view or modify data that is relevant to their role.

For example, HR users are allowed to access employee-related information, while Finance users can work with financial and billing data according to their permissions. This separation helps protect sensitive information and reduces the risk of unauthorised access. These controls are consistent with standard enterprise practices for systems that handle confidential organisational data.

In addition to access control, physical security is ensured through the data centre infrastructure where SAP servers are hosted. Backup systems and high availability configurations further strengthen data protection by ensuring that information can be recovered if system failures occur. Together, these measures help maintain the confidentiality, integrity, and availability of stored data.

8.4.5 Long-Term Storage

Long-term storage refers to the retention of historical data over extended periods for compliance, auditing, and analysis purposes. In SAP ERP, many types of records such as payroll data, financial transactions, and HR records must be stored for several years based on regulatory requirements.

The system is designed to support this long-term retention without affecting daily operations. Historical data remains accessible when needed for audits or reporting, while backup systems ensure that it is protected from loss. However, as the database continues to grow over time, proper capacity planning is necessary to maintain system performance and ensure that storage resources remain sufficient for future needs.

8.5 Feedback

8.5.1 What is Feedback?

Feedback refers to information generated by an information system that reflects its own performance, outputs, and operational behaviour. Unlike outputs that are directed to users or external stakeholders, feedback is mainly used internally by IT teams and management to assess system effectiveness and identify areas for improvement. It plays an important role in maintaining system stability and ensuring that performance issues are addressed early.

In the SAP ERP system at Pos Malaysia, feedback is mainly related to technical performance and system behaviour. It helps the IT team understand how the system is functioning under real operating conditions and supports decision-making related to system optimisation, resource allocation, and long-term planning.

8.5.2 Performance Monitoring

Performance monitoring is the main feedback mechanism used in the SAP environment. Based on the interview, the IT team continuously tracks several key indicators to evaluate system health. One of the most important metrics is CPU usage, which shows how much processing power is being used at a given time. When CPU usage becomes too high, it may indicate that the system is under heavy load, which can slow down processing and affect user experience.

Memory usage is also monitored to ensure that sufficient RAM is available for system processes. If memory becomes insufficient, the system may rely on slower storage-based processing, which reduces performance. In addition, IOPS (Input/Output Operations Per Second) is used to measure how efficiently the system reads and writes data to storage. High demand in this area can create bottlenecks that delay database activity.

Another key indicator is query response time, which reflects how quickly the system responds to user or system requests. Slow response times may suggest the need for database optimisation or improved system configuration. Together, these metrics provide a clear picture of system performance and help the IT team respond to potential issues before they affect operations.

8.5.3 Error Monitoring

Error monitoring provides feedback on system failures and processing issues. SAP automatically generates logs that record errors such as failed transactions, data validation issues, and interface communication problems. These logs are regularly reviewed by the IT team to identify recurring issues and determine their root causes.

This process is especially important because SAP at Pos Malaysia is connected to around 165 system interfaces. Each connection represents a possible point of failure, and any disruption can affect the flow of data between systems. By monitoring interface errors, the IT team can quickly detect problems and restore normal data exchange, reducing the risk of delays or data inconsistencies.

8.5.4 Capacity Planning

Capacity planning uses feedback from system performance to prepare for future infrastructure needs. The IT team analyses trends such as CPU usage, memory consumption, data growth, and user activity to estimate when system resources will need to be expanded. This allows upgrades to be planned in advance rather than reacting to performance issues after they occur.

With the SAP database already storing around 11 terabytes of data and continuing to grow, storage capacity is a major consideration. Capacity planning ensures that the system remains stable even as data volumes increase over time. This proactive approach helps maintain performance and prevents system limitations from affecting business operations.

8.5.5 Continuous Improvement and Performance Optimisation

Feedback from monitoring and error tracking is used to support continuous improvement of the SAP ERP system. When performance issues are identified, the IT team investigates the cause and applies corrective actions such as system tuning, database optimisation, or infrastructure adjustments. This helps improve overall system efficiency and stability.

Similarly, recurring errors are analysed to determine whether they originate from system configuration, user input, or integration issues. In some cases, improvements are made at the input stage to prevent the same errors from occurring again. Regular system updates and configuration changes are also part of this improvement cycle.

Overall, this continuous feedback loop ensures that the SAP system at Pos Malaysia evolves over time. It is not a static system but one that is actively monitored and improved to meet increasing operational demands and changing business requirements.

9.0 HARDWARE AND SOFTWARE CONFIGURATION

9.1 Hardware Configuration

Pos Malaysia is a postal and logistics provider. They handle massive volumes of data from customer delivery data to complex financial data on a daily basis. The firm uses two server infrastructure which consists of both cloud-based and physical (non-cloud) servers to

ensure these processes work properly. These servers are of their basic SAP ERP system, put in place around 2013 to handle high-volume operations such as processing up to 100,000 invoices and managing staff payroll centrally.

Pos Malaysia uses networking technology such as Local Area Networks (LAN) and Wide Area Networks (WAN) to ensure speedy information transfer among the various branches throughout the country. The network architecture plays an important role in real-time updates. For example, when a client delivers a box through Pos Laju, the system updates the delivery status promptly at all locations. This helps the team to coordinate better and improves the service quality.

Lastly, Pos Malaysia highlights the reliability of the system through High Availability (HA) system and backup servers as the system outage would be a huge business issue. All of these components work together to prevent data loss and provide a rapid backup if the primary server fails. This hardware combined with a good disaster recovery plan will allow the organization to continue its business and deliver uninterrupted customer service.

9.2 Software Configuration

Pos Malaysia's software setup is a crucial aspect of the company's ability to remain efficient and up to date. The main of their operations is the SAP ERP system first introduced in 2013. The software is centralized and connects the many departments such as Human Resources (HR), Finance etc to communicate data seamlessly rather than having separate disconnected files. The corporation can use this system to automate the routine duties such as tracking employee attendance, processing payroll and computing benefits like EPF and tax. This saves time and avoids errors which are common with manual work.

Pos Malaysia manages its huge volume of business data by storing it in Relational Database Management Systems (RDBMS) with SQL technology. These databases are meant to organize structured data, such as the 100,000 billing records and invoices managed by the Finance Department so they can be searched and retrieved promptly. The data is so organized that managers have real-time reports that lead to better, faster business decisions.

Another important element of the software configuration is how the many systems communicate with each other using Application Programming Interfaces (APIs). APIs allow other software direct access to the core SAP ERP system such as courier systems used by Pos Laju or Customer Relationship Management (CRM) systems. When logistics updates the status of a parcel, this status will automatically flow into the ERP for reporting. It's providing the latest picture of operations across the company.

Pos Malaysia's IT team make sure their system go smooth operations by constantly monitoring performance and making regular software updates and configuration adjustments.

They keep an eye on CPU utilization and response times. It is to prevent slowdowns during peak online usage. For instances, they must do continuous improvements are for important to upholding data security and supporting the company's commitment to providing reliable services to customers as it progresses on its digital transformation path.

10.0 INFORMATION SYSTEM PROBLEMS

10.1 Current Issues and Challenges

While Pos Malaysia's enterprise system is useful during the operation, it also has its challenges in managing such a technology. One of the major challenges is the very complex system. Many departments such as HR or Finance demand specialized technical and accounting expertise. Users cannot simply jump in. They typically need comprehensive training and even professional certification to use the system efficiently. This demands the human element of the system as challenging as the technology itself.

Another issue is slow system performance during peak periods. Pos Malaysia deals with a massive amount of data on a daily basis. It's including processing up to 100,000 invoices and managing records for over 200 different systems. When the workload is at its highest, the system can experience "overload" delays. For instance, if the Finance and HR departments are both running large reports or processing thousands of transactions at the same time, the response speed can slow down, which directly impacts employee productivity and the speed of service delivery.

The organization also struggles with the difficulties of maintenance and upgrades. Since SAP is a global, integrated solution, moving data or updating software is a delicate and expensive process. Any technical error during an update could potentially disrupt the entire business, affecting everything from parcel tracking at Pos Laju to employee payroll. This means the IT support team must be extremely careful with every configuration change to maintain business continuity.

Finally, cybersecurity and high costs are ongoing concerns. It is because the system stores sensitive financial, employee, and customer transaction data. It is a constant be target for cyber threats. Protecting this information requires active monitoring and high availability backup systems which adds to the already high cost of system licensing. To stay competitive and reliable, Pos Malaysia must continue to invest time and resources into overcoming these technical and security challenges.

10.2 Causes of the Problems

Pos Malaysia has a few technological problems with its information systems due a few factors. One of the main causes is the complexity of the SAP ERP system, which has been the basic platform of the firm since about 2013. The system is constructed on a series of interconnected modules like Human Resources, Finance, Billing etc. All of which rely on a common database. These different business functions are so interconnected that a technical problem in one module might readily impact the others. This "domino effect" makes it tougher for the IT support personnel to manage the system and fix problems. For the business, it's a continuing struggle to be able to scale with this complexity to over 200 sub-systems to monitor.

Another serious problem is the massive number of transactions a day. Pos Malaysia, one of the largest logistics providers, handles large volumes of data every day, including up to 100,000 invoices and thousands of payroll and delivery records. When the workload increases, the system can be overloaded which causes performance delays and delayed response times. These short-term blockages can be frustrating to staff and impede down service delivery. Moreover, the complex SAP platform is not accompanied by the specialist skills of the users. Employees need a unique mix of both IT competence and accounting understanding to use the system efficiently. Without regular training and professional certification, users are more prone to making errors that affect the accuracy of the company's data and overall productivity.

Lastly, the high maintenance and upgrade costs of the system are a major limitation. The cost of license and annual maintenance for running a big scale enterprise system like SAP is relatively high. Because these systems demand such a substantial financial investment, the business may sometimes be forced to delay essential software changes or have difficulty migrating data to newer, more efficient platforms. As the gadget matures these financial constraints can lead to more performance problems over time. In the end, the complexity of the systems, the large volumes of data, the requirement for highly skilled workers and the high operational costs combine to create a very demanding environment for Pos Malaysia's digital infrastructure.

10.3 Impact on Organizational Performance

One of the biggest problems is reduced operational efficiency. When the system experiences delays or becomes overloaded, employees find it much harder to finish their daily work on time. For example, processing up to 100,000 invoices or managing payroll and attendance can take much longer if the system is slow. These delays create a bottleneck that slows down the entire organization's workflow. In addition, these problems have a direct impact on the quality of the service.

Today's customers expect fast updates on their parcels, but if there are delays in the courier or CRM systems, the tracking information won't be available in real-time. These deficiencies in transparency can result in the loss of customer trust and reduced customer satisfaction. The managerial decision making process also gets hampered. Managers require accurate and real-time reports to prepare budgets and allocate resources, and when system errors happen, they are effectively without reliable data.

Pos Malaysia has implemented High Availability (HA) systems, backup servers and a disaster recovery plan to stop minor incident from becoming into a full-scale disaster. These tools and constant monitoring of CPU usage and error logs are critical to ensuring business goes on smoothly despite technical challenges. In the end, system failures will result in higher operating costs. Complex technical issues mean costly maintenance, troubleshooting and sometimes even specialist IT support.

11.0 PERFORMANCE OF THE INFORMATION SYSTEM

11.1 System Efficiency

First, system efficiency in Pos Laju Malaysia is automated processing. The system automatically handles payroll calculation, tax processing, billing activities and report generation. For instances, instead of manually processing the employee payroll records, HR staff can generate payroll result direct from the system within a minute. This can reduce errors and improves productivity.

Secondly, system ability to process a large transaction volume. According to the interview findings, the enterprise system can manage up to 100,000 invoices records. This capability allows Pos Malaysia to handle extensive business operations without increasing manpower.

Pos Malaysia also monitors system performance using indicator like CPU usage, query response time, Input and Output Operation Per Second (IOPS) and error rates. These monitoring activities help IT team identify performance issues faster and make sure smooth system operation.

Furthermore, the integration of a few departments into a single ERP platform reduces a duplication of work. For example, financial data entered by one department can be accessed directly by another department without requiring repeated data entry. This can improve information accuracy and time.

11.2 System Effectiveness

System effectiveness is simply a measure of how well an information system helps an organization reach its goals. For Pos Malaysia, the SAP ERP system is highly effective because it supports vital business functions by providing accurate data and making it easier for different departments to work together.

The best example of this success is business process integration. The system links Human Resources, Finance, and billing into one central platform. For instance, payroll data created by the HR team is automatically shared with the Finance department for accounting. This all-in-one approach keeps information consistent across the company and prevents the common mistakes that happen when data is handled manually.

The system also proven its worth through timely reporting. Reports such as payroll, tax records, and financial analyses are used by managers to make strategic, informed decisions. It is because the system can handle a massive workload processing up to 100,000 invoices. It provides real-time information that helps leaders evaluate the company's performance and plan for the future more confidently.

Finally, the system enhances operational efficiency by automating repetitive tasks. By automating workflows for tasks such as generating invoices, Pos Malaysia has reduced manual work and therefore operations are faster and with fewer human errors. Along with active monitoring tools that track system condition and errors, the SAP ERP system keeps Pos Malaysia running smoothly, coordinated and achieving its business objectives effectively.

11.3 User Satisfaction

To understand if an information system is truly successful, we have to look at user satisfaction. At Pos Malaysia, this depends on how well the SAP ERP system supports daily work and helps employees be more productive. It is because the system is designed to meet the needs of a large-scale enterprise, satisfaction is tied to its reliability and the "convenience factor" it brings to a busy workplace.

One of the biggest reasons staff find the system helpful is its centralized design. Instead of switching between many different applications, teams in Human Resources and

Finance can manage almost everything from tracking employee attendance and benefits to handling billing within one single platform.

Speed and accuracy of automation are also one of user satisfaction. By removing slow, manual accounting the system is now able to process and produce payroll reports much faster than previously. This is a huge win for employees as it enables them to stop doing boring repetitive paperwork and focus on more important tasks that actually add value to the company.

But it is not always easy to achieve a high satisfaction because the SAP platform is very complex. New users often find the system's many modules and technical features difficult to understand at first. To fix this, Pos Malaysia requires extensive training and professional certification. While this means employees have to work hard to learn the system, it eventually increases their confidence and makes them more skilled in using advanced technology

User satisfaction at Pos Malaysia remains positive. The benefits like remote access for employees and faster service for customers make the system a success. In the end, the massive gains in efficiency and coordination far outweigh the challenges of learning how to manage such a sophisticated system

11.4 Contribution to Organizational Operations

Pos Malaysia has been operating its core enterprise system using SAP ERP since 2013. It connects more than 200 systems of different departments and is the digital backbone of the firm. A single platform allows employees to effortlessly access and exchange information with each other, helping departments such as Human Resources, Finance and Operations work together more efficiently.

One of the system's greatest accomplishments has been the automation of daily business procedures. Tasks such as payroll, billing, invoicing and report preparation are done faster and with fewer errors. For instance, the system can handle up to 100,000 invoices, thus Pos Malaysia may execute a significant number of transactions without having to increase its workforce.

Moreover, the system facilitates communication by enabling real-time information sharing via integrated modules and APIs. Information like parcel tracking can be shared rapidly across departments. Enabling employees to respond quicker to customers' requests. Thus, SAP ERP has enhanced operational efficiency, decision making and enabled Pos Malaysia to provide faster and more reliable logistics services.

12.0 SWOT ANALYSIS OF THE INFORMATION SYSTEM

12.1 Strengths

The SAP ERP system at Pos Malaysia Berhad demonstrates several notable strengths that directly contribute to the organisation's operational effectiveness and data management capability.

The most significant strength of the system is its centralised and integrated architecture. Rather than relying on separate departmental tools that operate in isolation, SAP brings together the Human Resource and Finance functions into a single unified platform. This integration allows data to flow seamlessly between the two departments without the need for manual re-entry, thereby improving both accuracy and efficiency. For example, payroll data processed within the Human Capital Management (HCM) module is automatically reflected in the financial accounts managed by the Finance and Controlling (FI/CO) module, ensuring consistency across the organisation.

Another key strength is the system's capacity to support high transaction volumes. As revealed during the interview, the SAP ERP system processes payroll for approximately 15,000 employees and generates around 100,000 invoices per processing cycle. The system's ability to manage these volumes reliably and consistently demonstrates its suitability for a national-scale organisation such as Pos Malaysia. Additionally, the system maintains an overall database of approximately 11 terabytes, reflecting the scale and continuity of its data management operations.

The system also supports real-time monitoring and reporting, which is a considerable operational advantage. Through the SAP platform, managers can access up-to-date financial reports, cost centre summaries, accounts receivable ageing reports, and payroll records without depending on manual data compilation. This real-time visibility supports timely and well-informed decision-making at both the operational and strategic levels.

12.2 Weaknesses

Despite its many strengths, the SAP ERP system is not without its limitations, and several weaknesses have been identified through the findings of the interview conducted with Pos Malaysia.

One of the primary weaknesses is the complexity of the system and the steep learning curve associated with its use. The SAP ERP platform is not intuitive for general users, and employees are required to undergo formal training and obtain relevant certifications before they are able to operate their respective modules effectively. As noted during the interview, users of the Finance module need to possess accounting knowledge in addition to technical

IT proficiency. This requirement for specialised training increases the time and cost of onboarding new users and limits the accessibility of the system to a relatively small group of skilled personnel, which currently stands at approximately 200 active users.

Moreover, the high cost associated with SAP implementation, licensing, maintenance, and upgrades also represents a significant financial burden for the organisation. SAP systems are recognised globally as enterprise-level solutions that demand substantial investment, and Pos Malaysia must engage external SAP vendors for licensing agreements, technical support, and system updates. The reliance on external vendor relationships introduces an additional layer of dependency and cost that is not always predictable.

Lastly, despite having a team responsible for system administration, the IT team supporting the SAP ERP system is relatively small, comprising approximately eight individuals including internal staff, external vendors, and contractors. Given the scale of the system and the critical nature of the data it manages, the limited size of this team could pose a risk in scenarios requiring rapid response to system failures or security incidents.

12.3 Opportunities

One of the most significant opportunities is the potential for system upgrades and migration to more advanced versions of SAP. SAP S/4HANA, the latest generation of the SAP ERP platform, offers improved processing speed, enhanced in-memory computing capabilities, and a more user-friendly interface compared to older SAP versions. Migrating to SAP S/4HANA would allow Pos Malaysia to benefit from more powerful analytics, greater automation, and a simplified data model, all of which would strengthen the organisation's decision-making capabilities and operational efficiency.

Another opportunity is the expansion of automation within the ERP system. As technology advances, there is growing potential to integrate Artificial Intelligence (AI) and machine learning tools into the existing SAP platform to automate repetitive processes such as invoice matching, anomaly detection in financial data, and predictive workforce planning. Pos Malaysia has already expressed interest in adopting AI and data analytics in its logistics operations through its Annual Report 2023, and this strategic direction could naturally extend into the ERP environment as well.

Furthermore, improvements in reporting and business intelligence capabilities represent another avenue for development. By connecting the SAP ERP system more deeply with the Big Data Analytics platform, Pos Malaysia could generate more sophisticated insights from its HR and financial data, enabling better strategic planning and resource allocation across the organisation.

12.4 Threats

Several external threats could potentially affect the stability, security, and continuity of the SAP ERP system at Pos Malaysia Berhad.

One of the most pressing threats is the risk of system overload and performance degradation. Given the high volume of transactions that the SAP system processes on a regular basis which 100,000 invoices per cycle and payroll for 15,000 employees every month. Therefore, any significant increase in workload or inefficiency in system resource management could result in processing delays and reduced system responsiveness. As Pos Malaysia continues to grow its logistics and e-commerce operations, the demand placed on the ERP system is likely to increase accordingly.

Cybersecurity threats represent another major concern. As a publicly listed national organisation that manages sensitive employee data, financial records, and customer billing information, Pos Malaysia is a potential target for cyberattacks, including data breaches, ransomware, and unauthorised system access. Any compromise of the SAP system could have severe consequences for the organisation's financial reporting, regulatory compliance, and reputation.

The organisation's high dependency on SAP also introduces a significant operational risk. If the SAP vendor alter its licensing terms, discontinue support for the current version, or experience service disruptions, Pos Malaysia would face considerable challenges in maintaining its HR and Finance operations. The deep integration of SAP with 165 other systems means that any failure within the ERP platform could have a cascading effect on the entire information architecture of the organisation.

Finally, the risk of system crashes or data loss, while mitigated by the existing High Availability (HA) configurations and disaster recovery plans implemented by the IT team, remains a latent threat. The criticality of the data stored within the SAP system — including payroll records, tax documents, and financial statements — means that even a brief period of unplanned downtime could have significant operational and regulatory consequences.

13.0 GENERATE AND EVALUATE ALTERNATIVES

13.1 Alternative 1

The first alternative that we want to propose is that Pos Malaysia Berhad should develop and implement a comprehensive, formally structured Backup and Restore System together with a Disaster Recovery Plan (DRP) specifically designed for the SAP ERP environment. This alternative seeks to formalise, standardise, and significantly strengthen the organisation's ability to recover from system failures, data loss, or major disruptions in a timely and controlled manner.

The backup strategy under this alternative would be restructured into a tiered framework consisting of full backups, incremental backups, and differential backups scheduled at defined intervals throughout the day, week, and month. Backup data would be stored across multiple locations, including on-premises storage and the organisation's existing Amazon Web Services (AWS) cloud infrastructure, to ensure geographical redundancy. Encryption of all backup data would be mandated to protect the confidentiality of sensitive employee and financial records during storage and transmission.

The Disaster Recovery Plan itself would be formally documented and would cover detailed response procedures for a range of failure scenarios, including server hardware failure, database corruption, ransomware or cyberattack, natural disaster, and network outage. The plan would assign clear roles and responsibilities to members of the IT team and key departmental stakeholders, and would specify escalation procedures, communication protocols, and step-by-step system restoration sequences. Critically, regular disaster recovery drills would be a mandatory on a scheduled basis which will be set at minimum once every six months. This is to verify that backup restoration procedures work as intended and that all responsible personnel are familiar with their roles during a recovery event. The results of each drill would be formally recorded and used to refine and improve the plan over time.

13.2 Alternative 2

The second alternative does not support for a full system replacement or migration. Instead, it proposes that Pos Malaysia retains the existing SAP ERP system and addresses its current weaknesses by supplementing it with a dedicated cybersecurity infrastructure and a structured system performance management framework.

Under this alternative, Pos Malaysia recommends investing in a comprehensive cybersecurity enhancement programme that includes the deployment of advanced intrusion detection systems (IDS), Security Information and Event Management (SIEM) tools, and role-based access control (RBAC) auditing mechanisms specifically configured for the SAP environment. Regular penetration testing and vulnerability assessments would also be

scheduled to identify and address security gaps before they can be exploited. These measures would directly address the cybersecurity threats identified in the SWOT analysis.

Furthermore, a performance management system should be established to proactively monitor and optimise the SAP system's resource utilisation. This would involve deploying advanced monitoring tools to track CPU usage, memory consumption, IOPS metrics, and query response times against predefined performance limit. When performance issues are identified, specific optimisations such as database indexing, query optimisation, and load balancing can be implemented without requiring a full system overhaul. The IT team would also be expanded and formally structured to ensure adequate coverage for system support, incident response, and maintenance activities within the organisation.

This alternative also includes the development of a formal disaster recovery and business continuity plan that goes beyond the existing High Availability configuration, with regular testing of backup restoration procedures and the establishment of a secondary recovery site to ensure operational continuity in the event of a major system failure.

13.3 Evaluation of Alternatives

Both alternatives offer viable pathways to addressing the current limitations of the SAP ERP system, but each comes with a distinct set of trade-offs that must be carefully considered by the organisation.

Alternative 1, the implementation of a structured Backup, Restore, and Disaster Recovery Plan, addresses one of the most operationally critical vulnerabilities of the current SAP ERP environment. Given that the system manages approximately 11 terabytes of sensitive financial and HR data and is integrated with 165 system interfaces, the consequences of an unplanned system failure without a formalised recovery plan could severely impact operational downtime to irreversible data loss and regulatory non-compliance. Moreover, this alternative is relatively focused on scope and does not require replacing or significantly altering the core SAP system, which means implementation risk is comparatively low. However, formalising and maintaining a robust disaster recovery programme does require a meaningful investment of time, personnel, and infrastructure resources, particularly in establishing geographically redundant backup storage, conducting regular recovery drills, and training the IT team to execute recovery procedures effectively under pressure.

Alternative 2, the implementation of cybersecurity and performance management framework, is similarly non-disruptive and addresses a broader range of operational risks, including cybersecurity threats, system performance degradation, and the absence of proactive monitoring mechanisms. Since it does not require replacing the core ERP system, the risk of operational interruption is minimal. However, the limitation of this alternative is that

while it strengthens the protective and monitoring layer around the SAP system, it does not specifically address the organisation's ability to recover effectively from a major system failure or data loss event. In conclusion, this alternative leaves a significant gap in the organisation's overall system resilience strategy.

13.4 Comparison of Alternatives

CRITERIA	ALTERNATIVE 1	ALTERNATIVE 2
Cost	Moderate	Moderate
Implementation Risk	Low	Low
Disruption to Overall System	Minimal	Minimal
Addresses The Data Loss Risk	Fully	Partially
Addresses The Cybersecurity Risk	Partially	Fully
Improves Performance Monitoring	No	Yes

14.0 RECOMMENDATIONS AND ACTION PLAN

14.1 Recommended Solution

After reviewing the SWOT analysis and carefully analyse the two proposed alternatives, it best recommends that Pos Malaysia Berhad moves forward with implementing both alternatives at the same time, as they are designed to work hand in hand rather than as separate implementation. The first alternative, which focuses on building a proper Backup, Restore, and Disaster Recovery Plan, should be carried out together with the second alternative, which involves strengthening the organisation's cybersecurity controls and performance monitoring practices. Therefore, when combined, these two initiatives will support the most critical vulnerabilities of the SAP ERP system and the need to recover the system quickly when the system goes wrong, and the need to prevent problems from happening in the first place.

Furthermore, beyond these two core alternatives, it also recommends four additional improvements that address the human and day-to-day operational side of the SAP ERP system. These include improving user training to be specific to each SAP module, increasing the use of automation in reporting processes, enhancing how the system performs technically, and making the system easier and more comfortable to use for those who rely on it daily. Taken together with the two main alternatives, these improvements form a well-rounded plan that addresses both the technical and human dimensions of the system.

14.2 Justification of Recommendation

The decision to recommend both alternatives together comes from the understanding that system security and the ability to recover from failures are both important to ensure the effectiveness of the system. Having strong cybersecurity measures in place is important, but if the organisation lacks a proper recovery plan, a single serious incident such as a data breach or server failure could still result in significant data loss and prolonged disruption. On the other hand, having a well-prepared disaster recovery plan means very little if the system remains poorly protected and vulnerable to threats that could trigger a recovery situation in the first place. Furthermore, considering that the SAP ERP system is responsible for processing payroll for 15,000 employees every month, handling approximately 100,000 invoices each cycle, and maintaining financial records that are subject to regulatory reporting requirements to Bursa Malaysia, making mistakes in either of these areas can have serious consequences. Especially for an organisation of Pos Malaysia's size and national responsibility, addressing both dimensions together is not merely advisable, it is necessary.

Furthermore, both alternatives are comparable in terms of cost, feasibility, and implementation timeline, making it practical for Pos Malaysia to pursue them simultaneously without placing an unreasonable burden on its IT team or financial resources. The combined implementation of these two initiatives represents the most prudent and comprehensive approach to safeguarding the long-term reliability and integrity of the SAP ERP system.

14.3 Action Plan

The following action plan outlines the key steps and timeline for implementing both recommended alternatives simultaneously.

Phase 1 — Disaster Recovery Planning and Backup Infrastructure (Months 1–3)

In the first month, a Business Impact Analysis (BIA) should be conducted to assess the criticality of each SAP-supported business function and establish formal Recovery Time Objectives (RTOs) and Recovery Point Objectives (RPOs). During months two and three, a tiered backup strategy should be designed and implemented, incorporating full, incremental, and differential backups stored across both on-premises infrastructure and the AWS cloud environment. All backup data should be encrypted to protect sensitive HR and financial records. A formally documented Disaster Recovery Plan covering a range of failure scenarios should be completed by the end of this phase, with clearly assigned roles, escalation procedures, and step-by-step restoration sequences.

Phase 2 — Cybersecurity Enhancement and Performance Monitoring (Months 1–3, concurrent)

Running concurrently with Phase 1, a comprehensive cybersecurity audit of the SAP ERP system should be initiated in the first month to identify vulnerabilities in access controls, data encryption, and system configuration. Based on the audit findings, SIEM tools and strengthened role-based access control (RBAC) configurations should be deployed during months two and three. Advanced performance monitoring tools should also be configured during this phase to establish baselines for CPU utilisation, memory usage, IOPS, and query response times, with automated alerts set for critical threshold breaches.

Phase 3 — Testing, Validation, and Continuous Improvement (Months 4–6)

During this phase, the first formal disaster recovery drill should be conducted to test backup restoration procedures and verify the effectiveness of the documented DRP. The results of the drill should be reviewed, and any identified gaps should be addressed through plan revisions. Quarterly penetration testing of the SAP environment should be scheduled and initiated. A formal IT team expansion and capacity plan should also be finalised during this phase to ensure the organisation has adequate human resources to sustain both initiatives on an ongoing basis.

14.4 Expected Benefits

The combined implementation of both recommended alternatives is expected to provide several significant and immediate benefits for Pos Malaysia Berhad's system.

First, in terms of disaster recovery readiness, the implementation of a structured Backup, Restore, and Disaster Recovery Plan will ensure that the organisation is prepared to respond and recover from system failures in a controlled and timely manner. Furthermore, regular recovery drills will build the IT team's confidence and competence in executing recovery procedures, reducing recovery time in actual emergency situations.

Next, in terms of cybersecurity and performance, the enhancements introduced previously will reduce the organisation's exposure to data breaches and system performance degradation. Furthermore, proactive monitoring will also enable the IT team to detect and resolve emerging issues before they escalate into serious incidents, improving the overall stability and reliability of the SAP platform.

To sum up, the implementation of both alternatives will strengthen the organisation's overall information system resilience, protect the integrity of its sensitive data, reduce operational risk, and enhance Pos Malaysia's ability to meet its regulatory and reporting obligations with confidence.

15.0 CONCLUSION

To conclude, this project was conducted as part of the IMS552 Information Systems Management course requirement and aimed to examine how information systems are managed and utilised within a real organisational environment. Therefore, through an interview with a representative of Pos Malaysia Berhad, this study successfully explored the organisation's background, its business operations, and the range of information systems that support its national postal and logistics functions. Furthermore, six key information systems were identified during the interview, and the SAP Enterprise Resource Planning (ERP) system was selected as the focal system for in-depth analysis.

Moreover, in response to the findings of the SWOT analysis, two strategic alternatives were proposed and evaluated. The evaluation concluded that neither alternative can be effective if it were implemented independently. Furthermore, all five objectives established for this project also have been successfully achieved. The organisational background and structure of Pos Malaysia Berhad were thoroughly examined, the information systems in use were investigated and documented, the SAP ERP system's functions, components, and integration architecture were analysed in detail, the SWOT analysis provided a well-rounded evaluation of the system's strengths and vulnerabilities, and the recommendations and action plan offered a practical, time-bound roadmap for addressing the identified limitations.

In conclusion, the SAP ERP system is a critical and highly effective information system that has made a substantial contribution to the operational efficiency, data accuracy, and decision-making capability of Pos Malaysia Berhad since its implementation in approximately 2013. However, as the organisation continues to expand its logistics and e-commerce operations and as the volume and sensitivity of data managed within the system continue to grow, the need to proactively address its vulnerabilities becomes very important.

REFERENCES

10 best ERP Software in Malaysia (2026 reviewed). (2025).
<https://www.hashmicro.com/my/blog/best-erp-software/>

Annual report 2023 - cloudfront.net. (2023).
https://d38eiojdjahdit.cloudfront.net/Annual_Report_2023_6264cf6c09.pdf

Pos Malaysia | your trusted Malaysian Courier & Logistics Provider | Pos Laju. (2026).
<https://www.pos.com.my/about-us/leadership/senior-management-team>

Pos Malaysia | your trusted Malaysian Courier & Logistics Provider | Pos Laju. (2026).
<https://www.pos.com.my/about-us/bod>

SAP business one ERP Solutions malaysia - selatan technology. (2024).
<https://www.selatangroup.com/sap-business-one/>

The 7 best ERP systems for growing businesses in Malaysia. (2025).
<https://threadgoldconsulting.com/insights/unlocking-business-success-with-erp-systems-in-malaysia>

What is a POS system? (2023). <https://www.oracle.com/food-beverage/what-is-pos/>

What is Big Data? | IBM. (2023). <https://www.ibm.com/think/topics/big-data>

What is CRM (Customer Relationship Management)? | salesforce. (2023).
<https://www.salesforce.com/crm/what-is-crm/>

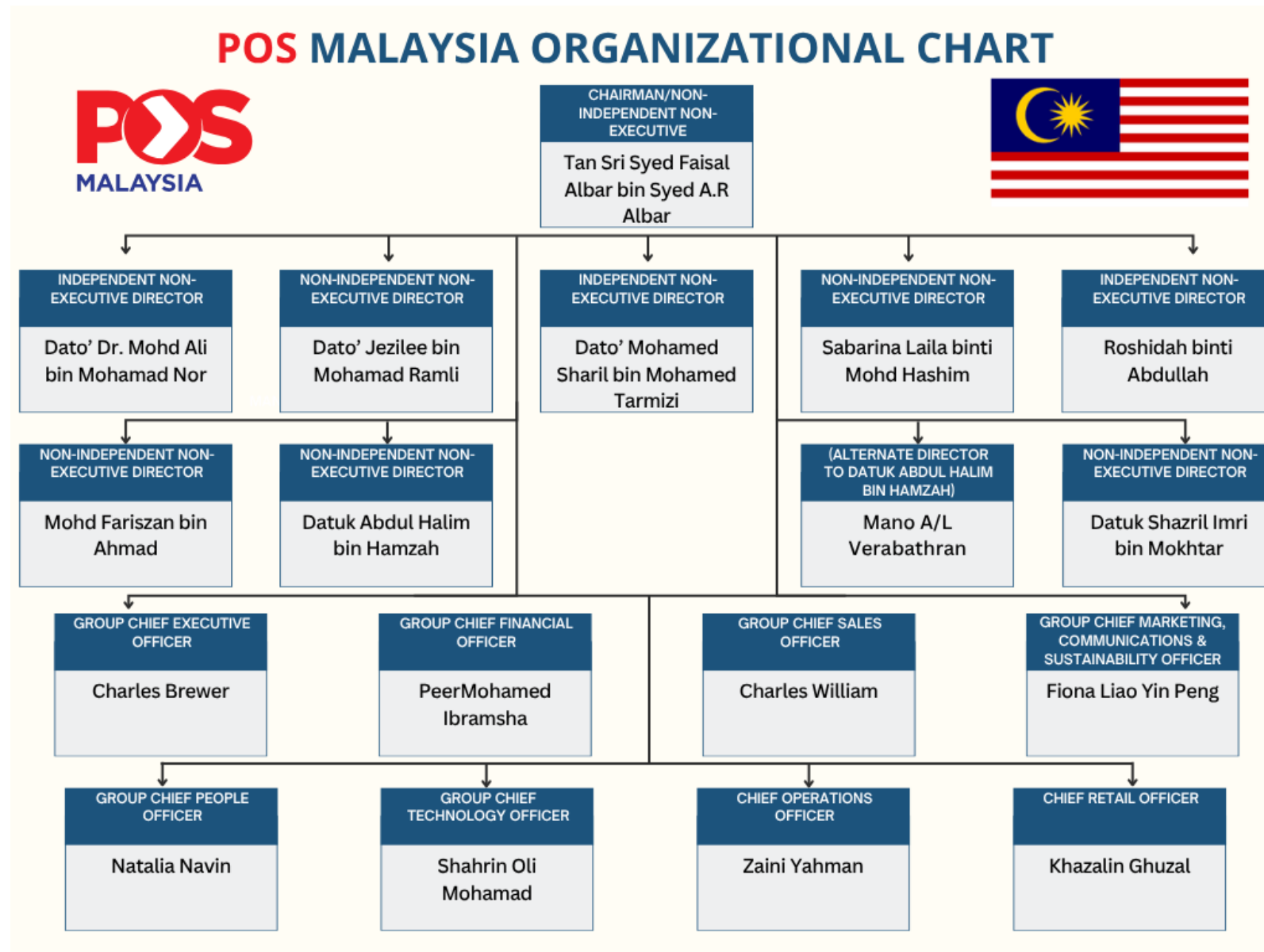
APPENDICES

Appendix A: Interview Questions

Appendix B: Interview Photos



Appendix C: Organization Structure Chart



Appendix D: Information System Screenshot

